2-11. A 100-MVA 230/115-kV Δ-Y three-phase power transformer has a per-unit resistance of 0.015 pu and a per-unit reactance of 0.06 pu. The excitation branch elements are $R_C = 100$ pu and $X_M = 20$ pu.

(a) If this transformer supplies a load of 80 MVA at 0.8 PF lagging, draw the phasor diagram of one phase of the transformer.

(b) What is the voltage regulation of the transformer bank under these conditions?

(c) Sketch the equivalent circuit referred to the low-voltage side of one phase of this transformer. Calculate all the transformer impedances referred to the low-voltage side.

(d) Determine the losses in the transformer and the efficiency of the transformer under the conditions of part *(b)*.

SOLUTION

(a) The transformer supplies a load of 80 MVA at 0.8 PF lagging. Therefore, the secondary line current of the transformer is

$$I_{LS} = \frac{S}{\sqrt{3}V_{LS}} = \frac{80{,}000{,}000 \text{ VA}}{\sqrt{3}(115{,}000 \text{ V})} = 402 \text{ A}$$

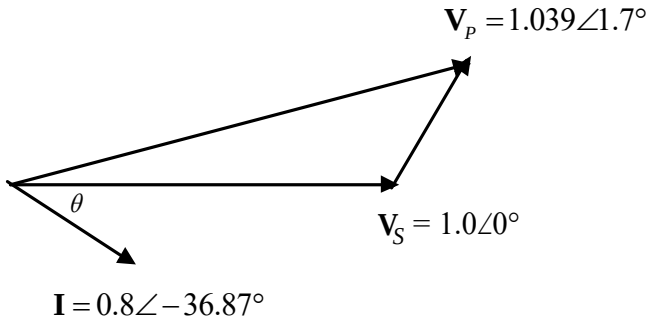
The base apparent power is $S_{\text{base}} = 100$ MVA, and the base line voltage on the secondary side is $V_{LS,\text{base}} = 115$ kV, so the base value of the secondary line current is

$$I_{LS,\text{base}} = \frac{S_{\text{base}}}{\sqrt{3}V_{LS,\text{base}}} = \frac{100{,}000{,}000 \text{ VA}}{\sqrt{3}(115{,}000 \text{ V})} = 502 \text{ A}$$

so the per-unit secondary current is

$$\mathbf{I}_{LS,\text{pu}} = \frac{I_{LS}}{I_{LS,\text{pu}}} = \frac{402 \text{ A}}{502 \text{ A}} \angle \cos^{-1}(0.8) = 0.8 \angle -36.87°$$

The per-unit phasor diagram is shown below:



(b) The per-unit primary voltage of this transformer is

$$\mathbf{V}_P = \mathbf{V}_S + \mathbf{I}\, Z_{\text{EQ}} = 1.0 \angle 0° + (0.8 \angle -36.87°)(0.015 + j0.06) = 1.039 \angle 1.7°$$

and the voltage regulation is

$$\text{VR} = \frac{1.039 - 1.0}{1.0} \times 100\% = 3.9\%$$

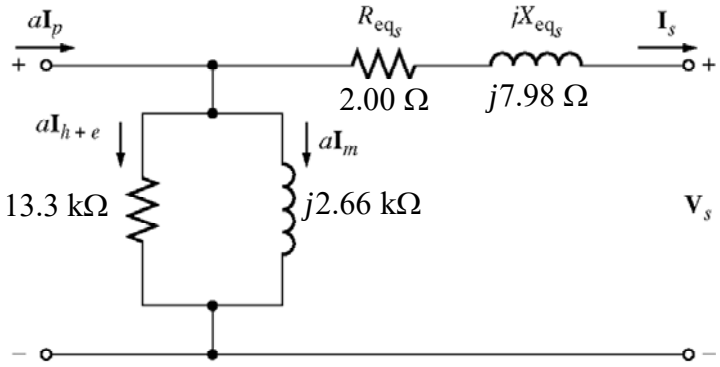
(c) The secondary side of this transfer is Y-connected, so the base phase voltage of the low voltage (secondary) side of this transformer is:

$$V_{\phi S,\text{base}} = \frac{V_{LS,\text{base}}}{\sqrt{3}} = \frac{115 \text{ kV}}{\sqrt{3}} = 66.4 \text{ kV}$$

The base impedance of the transformer referred to the low-voltage side is:

$$Z_{\text{base}} = \frac{3\, V_{\phi S,\text{base}}{}^2}{S_{\text{base}}} = \frac{3(66.4 \text{ kV})^2}{100 \text{ MVA}} = 133 \ \Omega$$

Each per-unit impedance is converted to actual ohms referred to the low-voltage side by multiplying it by this base impedance. The resulting equivalent circuit is shown below:



$$R_{\text{EQ,S}} = (0.015)(133\ \Omega) = 2.00\ \Omega$$
$$X_{\text{EQ,S}} = (0.06)(133\ \Omega) = 7.98\ \Omega$$
$$R_C = (100)(133\ \Omega) = 13.3\ \text{k}\Omega$$
$$X_M = (20)(133\ \Omega) = 2.66\ \text{k}\Omega$$

(d) The per-unit losses in the series resistance are

$$P_{EQ} = I^2 R_{EQ} = (0.8)^2 (0.015) = 0.0096\ \text{pu}$$

and the actual losses in the series resistance are:

$$P_{EQ} = S_{\text{base}} P_{EQ,\text{pu}} = (0.0096\ \text{pu})(100\ \text{MVA}) = 0.96\ \text{MW}$$

The per-unit losses in the excitation branch are:

$$P_{EX,\text{pu}} = \frac{V^2}{R_{EX}} = \frac{(1.039)^2}{100} = 0.0108\ \text{pu}$$

and the actual losses in the excitation branch are:

$$P_{EX} = S_{\text{base}} P_{EX,\text{pu}} = (0.0108\ \text{pu})(100\ \text{MVA}) = 1.08\ \text{MW}$$

The per-unit power supplied to the load

$$P_{\text{load.pu}} = \frac{P_{\text{load.}}}{S_{\text{base}}} = \frac{80\ \text{MW}}{100\ \text{MVA}} = 0.80\ \text{pu}$$

Therefore, the transformer's efficiency is

$$\eta = \frac{P_{\text{OUT}}}{P_{\text{IN}}} \times 100\% = \frac{0.80}{0.80 + 0.0096 + 0.0108} \times 100\% = 97.5\%$$

2-15. An autotransformer is used to connect a 12.6-kV distribution line to a 13.8-kV distribution line. It must be capable of handling 2000 kVA. There are three phases, connected Y-Y with their neutrals solidly grounded.

(a) What must the N_C / N_{SE} turns ratio be to accomplish this connection?

(b) How much apparent power must the windings of each autotransformer handle?

(c) What is the power advantage of this autotransformer system?

(d) If one of the autotransformers were reconnected as an ordinary transformer, what would its ratings be?

SOLUTION

(a) The transformer is connected Y-Y, so the primary and secondary phase voltages are the line voltages divided by $\sqrt{3}$. The turns ratio of each autotransformer is given by

$$\frac{V_H}{V_L} = \frac{N_C + N_{SE}}{N_C} = \frac{13.8 \text{ kV}/\sqrt{3}}{12.6 \text{ kV}/\sqrt{3}}$$

$$12.6\, N_C + 12.6\, N_{SE} = 13.8\, N_C$$

$$12.6\, N_{SE} = 1.2\, N_C$$

Therefore, $N_C / N_{SE} = 12.6 / 1.2 = 10.5$.

(b) The power advantage of this autotransformer is

$$\frac{S_{IO}}{S_W} = \frac{N_C + N_{SE}}{N_C} = \frac{N_C + 10.5 N_C}{N_C} = 11.5$$

so 1/11.5 of the power in each transformer goes through the windings. Since 1/3 of the total power is associated with each phase, the windings in each autotransformer must handle

$$S_W = \frac{2000 \text{ kVA}}{(3)(10.5)} = 63.5 \text{ kVA}$$

(c) As determined in (b), the power advantage of this autotransformer system is 11.5.

(d) The voltages across each phase of the autotransformer are $13.8/\sqrt{3}$ = 7967 V and $12.6/\sqrt{3}$ = 7275 V. The voltage across the common winding (N_C) is 7275 kV, and the voltage across the series winding (N_{SE}) is 7967 kV – 7275 kV = 692 V. Therefore, a single phase of the autotransformer connected as an ordinary transformer would be rated at 7275/692 V and 63.5 kVA.

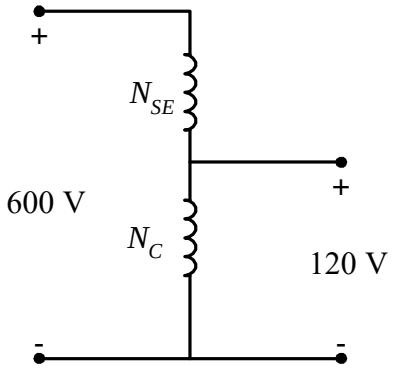
2-17. A 10-kVA 480/120-V conventional transformer is to be used to supply power from a 600-V source to a 120-V load. Consider the transformer to be ideal, and assume that all insulation can handle 600 V.

(a) Sketch the transformer connection that will do the required job.

(b) Find the kilovoltampere rating of the transformer in the configuration.

(c) Find the maximum primary and secondary currents under these conditions.

SOLUTION *(a)* For this configuration, the common winding must be the *smaller* of the two windings, and $N_{\text{SE}} = 4N_C$. The transformer connection is shown below:



(b) The kVA rating of the autotransformer can be found from the equation

$$S_{\text{IO}} = \frac{N_{\text{SE}} + N_C}{N_{\text{SE}}} S_W = \frac{4N_C + N_C}{4N_C}(10 \text{ kVA}) = 12.5 \text{ kVA}$$

(c) The maximum primary current for this configuration will be

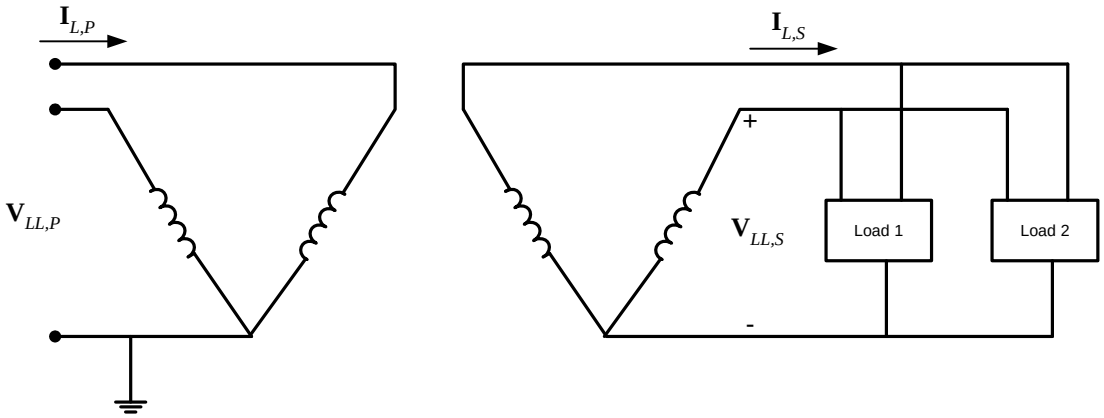
$$I_P = \frac{S}{V_P} = \frac{12{,}500 \text{ VA}}{600 \text{ V}} = 20.83 \text{ A}$$

and the maximum secondary current is

$$I_S = \frac{S}{V_S} = \frac{12{,}500 \text{ VA}}{120 \text{ V}} = 104 \text{ A}$$

2-19. Two phases of a 14.4-kV three-phase distribution line serve a remote rural road (the neutral is also available). A farmer along the road has a 480 V feeder supplying 200 kW at 0.85 PF lagging of three-phase loads, plus 60 kW at 0.9 PF lagging of single-phase loads. The single-phase loads are distributed evenly among the three phases. Assuming that the open-Y-open-Δ connection is used to supply power to his farm, find the voltages and currents in each of the two transformers. Also find the real and reactive powers supplied by each transformer. Assume that the transformers are ideal. What is the minimum required kVA rating of each transformer?

SOLUTION The farmer's power system is illustrated below:



The loads on each phase are balanced, and the total load is found as:

$$P_1 = 200 \text{ kW}$$

$$Q_1 = P_1 \tan\theta = (200 \text{ kW}) \tan\left[\cos^{-1}(0.85)\right] = 124 \text{ kvar}$$

$$P_2 = 60 \text{ kW}$$

$$Q_2 = P_2 \tan\theta = (60 \text{ kW}) \tan\left[\cos^{-1}(0.9)\right] = 29 \text{ kvar}$$

$$P_{\text{TOT}} = 260 \text{ kW}$$

$$Q_{\text{TOT}} = 153 \text{ kvar}$$

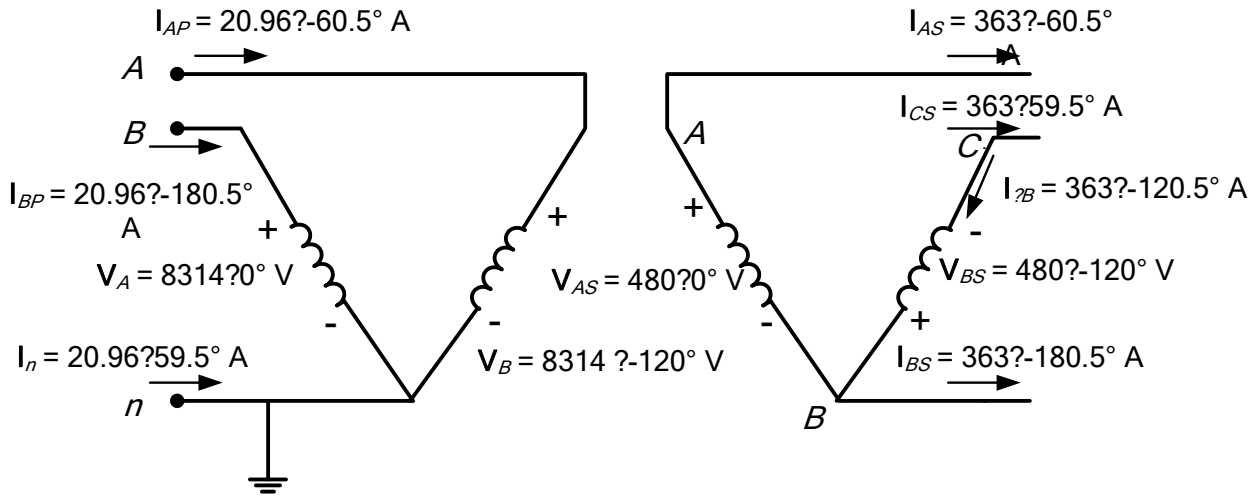
$$\text{PF} = \cos\left[\tan^{-1}\frac{Q_{\text{TOT}}}{P_{\text{TOT}}}\right] = \cos\left[\tan^{-1}\frac{153 \text{ kvar}}{260 \text{ kW}}\right] = 0.862 \text{ lagging}$$

The line current on the secondary side of the transformer bank is

$$I_{LS} = \frac{P_{\text{TOT}}}{\sqrt{3}\, V_{LS}\, \text{PF}} = \frac{260 \text{ kW}}{\sqrt{3}(480 \text{ V})(0.862)} = 363 \text{ A}$$

The open-Y—open Δ connection is shown below. From the figure, it is obvious that the secondary voltage across the transformer is 480 V, and the secondary current in each transformer is 246 A. The primary voltages and currents are given by the transformer turns ratios to be 7967 V and 14.8 A, respectively. If the voltage of phase *A* of the primary side is arbitrarily taken as an angle of 0°, then the voltage of phase *B* will be at an angle of –120°, and the voltages of phases *A* and *B* on the secondary side will be $\mathbf{V}_{AS} = 480\angle 0° \text{ V}$ and $\mathbf{V}_{BS} = 480\angle -120° \text{ V}$ respectively.

Note that line currents are shifted by 30° due to the difference between line and phase quantities, and by a further 30.5° due to the power factor of the load.



The real and reactive powers supplied by each transformer are calculated below:

$$P_A = V_{AS} I_A \cos\theta = (480\text{ V})(363\text{ A})\cos\left[0° - (-60.5°)\right] = 85.8\text{ kW}$$

$$Q_A = V_{AS} I_A \cos\theta = (480\text{ V})(363\text{ A})\sin\left[0° - (-60.5°)\right] = 151.7\text{ kvar}$$

$$P_B = V_{BS} I_{\phi B} \cos\theta = (480\text{ V})(363\text{ A})\cos\left[-120° - (-120.5°)\right] = 174.2\text{ kW}$$

$$Q_B = V_{BS} I_{\phi B} \sin\theta = (480\text{ V})(363\text{ A})\sin\left[-120° - (-120.5°)\right] = 1.5\text{ kvar}$$

Notice that the real and reactive powers supplied by the two transformers are radically different, put the apparent power supplied by each transformer is the same. Also, notice that the total power $P_A + P_B$ supplied by the transformers is equal to the power consumed by the loads (within roundoff error), while the total reactive power $Q_A + Q_B$ supplied by the transformers is equal to the reactive power consumed by the loads.

2-20. A 50-kVA 20,000/480-V 60-Hz single-phase distribution transformer is tested with the following results:

Open-circuit test (measured from secondary side)	Short-circuit test (measured from primary side)
V_{OC} = 480 V	V_{SC} = 1130 V
I_{OC} = 4.1 A	I_{SC} = 1.30 A
P_{OC} = 620 W	P_{SC} = 550 W

(a) Find the per-unit equivalent circuit for this transformer at 60 Hz.

(b) What is the efficiency of the transformer at rated conditions and unity power factor? What is the voltage regulation at those conditions?

(c) What would the ratings of this transformer be if it were operated on a 50-Hz power system?

(d) Sketch the equivalent circuit of this transformer referred to the primary side *if it is operating at 50 Hz.*

(e) What is the efficiency of the transformer at rated conditions on a 50 Hz power system, with unity power factor? What is the voltage regulation at those conditions?

(f) How does the efficiency of a transformer at rated conditions and 60 Hz compare to the same physical device running a 50 Hz?

SOLUTION

(a) The base impedance of this transformer referred to the primary side is

$$Z_{\text{base},P} = \frac{(V_P)^2}{S} = \frac{(20{,}000\ \text{V})^2}{50\ \text{kVA}} = 8\ \text{k}\Omega$$

The base impedance of this transformer referred to the secondary side is

$$Z_{\text{base},S} = \frac{(V_S)^2}{S} = \frac{(480\ \text{V})^2}{50\ \text{kVA}} = 4.608\ \Omega$$

The open circuit test yields the values for the excitation branch (referred to the *secondary* side):

$$|Y_{EX}| = \frac{I_{\phi,OC}}{V_{\phi,OC}} = \frac{4.10\ \text{A}}{480\ \text{V}} = 0.00854\ \text{S}$$

$$\theta = -\cos^{-1}\left(\frac{P_{OC}}{V_{OC}\ I_{OC}}\right) = -\cos^{-1}\left(\frac{620\ \text{W}}{(480\ \text{V})(4.1\ \text{A})}\right) = -71.6°$$

$$Y_{EX} = G_C - jB_M = 0.00854\angle -71.6° = 0.00270 - j0.00810$$

$$R_C = 1/G_C = 370\ \Omega$$

$$X_M = 1/B_M = 123\ \Omega$$

The excitation branch elements can be expressed in per-unit as

$$R_C = \frac{370\ \Omega}{4.608\ \Omega} = 80.3\ \text{pu} \qquad X_M = \frac{123\ \Omega}{4.608\ \Omega} = 26.7\ \text{pu}$$

The short circuit test yields the values for the series impedances (referred to the *primary* side):

$$|Z_{EQ}| = \frac{V_{SC}}{I_{SC}} = \frac{1130\ \text{V}}{1.30\ \text{A}} = 869\ \Omega$$

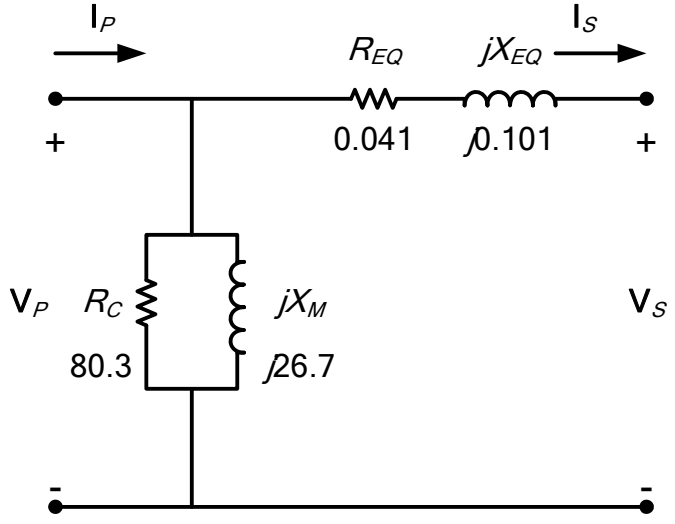
$$\theta = \cos^{-1}\left(\frac{P_{SC}}{V_{SC}\ I_{SC}}\right) = \cos^{-1}\left(\frac{550\ \text{W}}{(1130\ \text{V})(1.30\ \text{A})}\right) = 68.0°$$

$$Z_{EQ} = R_{EQ} + jX_{EQ} = 869\angle 68° = 326 + j806\ \Omega$$

The resulting per-unit impedances are

$$R_{EQ} = \frac{326\ \Omega}{8{,}000\ \Omega} = 0.041\ \text{pu} \qquad X_{EQ} = \frac{806\ \Omega}{8{,}000\ \Omega} = 0.101\ \text{pu}$$

The per-unit equivalent circuit is



(b) The per-unit primary voltage at rated conditions and unity power factor is

$$\mathbf{V}_P = \mathbf{V}_S + \mathbf{I}_S Z_{\text{EQ}}$$
$$\mathbf{V}_P = 1\angle 0° \text{ V} + (1\angle 0°)(0.041 + j0.101\ \Omega) = 1.046\angle 5.54° \text{ pu}$$

The per-unit power consumed by R_{EQ} is

$$P_{\text{EQ}} = I^2 R = (1 \text{ pu})^2 (0.041 \text{ pu}) = 0.041 \text{ pu}$$

The per-unit power consumed by R_C is

$$P_{\text{C}} = \frac{V_P{}^2}{R_C} = \frac{(1.046)^2}{80.3} = 0.0136 \text{ pu}$$

Therefore the efficiency of this transformer at rated load and unity power factor is

$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100\% = \frac{P_{\text{out}}}{P_{\text{out}} + P_{\text{EQ}} + P_C} \times 100\% = \frac{1.00}{1.00 + 0.041 + 0.0136} \times 100\% = 94.8\%$$

and the voltage regulation is

$$\text{VR} = \frac{1.046 - 1.00}{1.00} \times 100\% = 4.6\%$$

(c) The voltage and apparent power ratings of this transformer must be reduced in direct proportion to the decrease in frequency in order to avoid flux saturation effects in the core. At 50 Hz, the ratings are

$$S_{\text{rated}} = \frac{50 \text{ Hz}}{60 \text{ Hz}}(50 \text{ kVA}) = 41.7 \text{ kVA}$$

$$V_{P,\text{rated}} = \frac{50 \text{ Hz}}{60 \text{ Hz}}(20{,}000 \text{ V}) = 16{,}667 \text{ kV}$$

$$V_{S,\text{rated}} = \frac{50 \text{ Hz}}{60 \text{ Hz}}(480 \text{ V}) = 400 \text{ V}$$

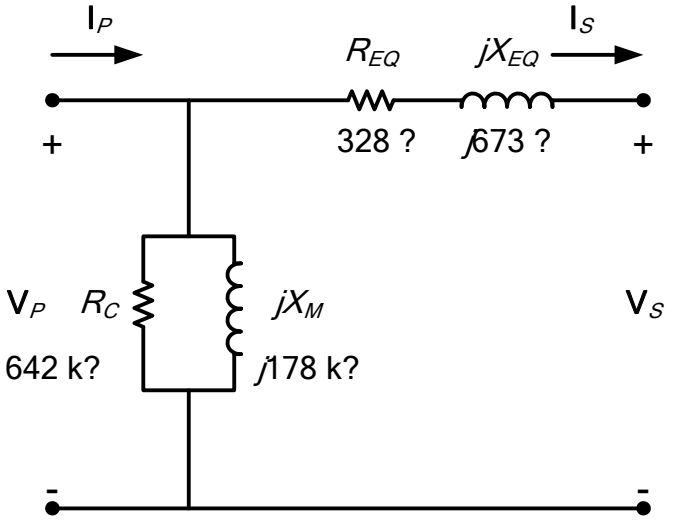
(d) The transformer parameters referred to the primary side *at 60 Hz* are:

$$R_C = Z_{\text{base}} R_{C,\text{pu}} = (8 \text{ k}\Omega)(80.3) = 642 \text{ k}\Omega$$
$$X_M = Z_{\text{base}} X_{M,\text{pu}} = (8 \text{ k}\Omega)(26.7) = 214 \text{ k}\Omega$$
$$R_{\text{EQ}} = Z_{\text{base}} R_{\text{EQ,pu}} = (8 \text{ k}\Omega)(0.041) = 328\ \Omega$$
$$X_{\text{EQ}} = Z_{\text{base}} X_{\text{EQ,pu}} = (8 \text{ k}\Omega)(0.101) = 808\ \Omega$$

At 50 Hz, the resistance will be unaffected but the reactances are reduced in direct proportion to the decrease in frequency. At 50 Hz, the reactances are

$$X_M = \left(\frac{50 \text{ Hz}}{60 \text{ Hz}}\right)(214 \text{ k}\Omega) = 178 \text{ k}\Omega$$
$$X_{\text{EQ}} = \left(\frac{50 \text{ Hz}}{60 \text{ Hz}}\right)(808\ \Omega) = \qquad \Omega$$

The resulting equivalent circuit referred to the primary at 50 Hz is shown below:



(e) The base impedance of this transformer *at 50 Hz* referred to the primary side is

$$Z_{\text{base},P} = \frac{(V_P)^2}{S} = \frac{(16{,}667\ \text{V})^2}{41.7\ \text{kVA}} = 6.66\ \text{k}\Omega$$

The base impedance of this transformer *at 50 Hz* referred to the secondary side is

$$Z_{\text{base},S} = \frac{(V_S)^2}{S} = \frac{(400\ \text{V})^2}{41.7\ \text{kVA}} = 3.837\ \Omega$$

The excitation branch elements can be expressed in per-unit as

$$R_C = \frac{642\ \text{k}\Omega}{6.66\ \text{k}\Omega} = 96.4\ \text{pu} \qquad X_M = \frac{178\ \text{k}\Omega}{6.66\ \text{k}\Omega} = 26.7\ \text{pu}$$

The series impedances can be expressed in per-unit as

$$R_{EQ} = \frac{328\ \Omega}{6.66\ \text{k}\Omega} = 0.0492\ \text{pu} \qquad X_{EQ} = \frac{673\ \Omega}{6.66\ \text{k}\Omega} = 0.101\ \text{pu}$$

The per-unit primary voltage at rated conditions and unity power factor is

$$\mathbf{V}_P = \mathbf{V}_S + \mathbf{I}_S Z_{\text{EQ}}$$

$$\mathbf{V}_P = 1\angle 0°\ \text{V} + (1\angle 0°)(0.0492 + j0.101\ \Omega) = 1.054\angle 5.49°\ \text{pu}$$

The per-unit power consumed by R_{EQ} is

$$P_{\text{EQ}} = I^2 R = (1\ \text{pu})^2\,(0.0492\ \text{pu}) = 0.0492\ \text{pu}$$

The per-unit power consumed by R_C is

$$P_{\text{C}} = \frac{V_P{}^2}{R_C} = \frac{(1.054)^2}{96.4} = 0.0115\ \text{pu}$$

Therefore the efficiency of this transformer at rated load and unity power factor is

$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100\% = \frac{P_{\text{out}}}{P_{\text{out}} + P_{\text{EQ}} + P_C} \times 100\% = \frac{1.00}{1.00 + 0.0492 + 0.0115} \times 100\% = 94.3\%$$

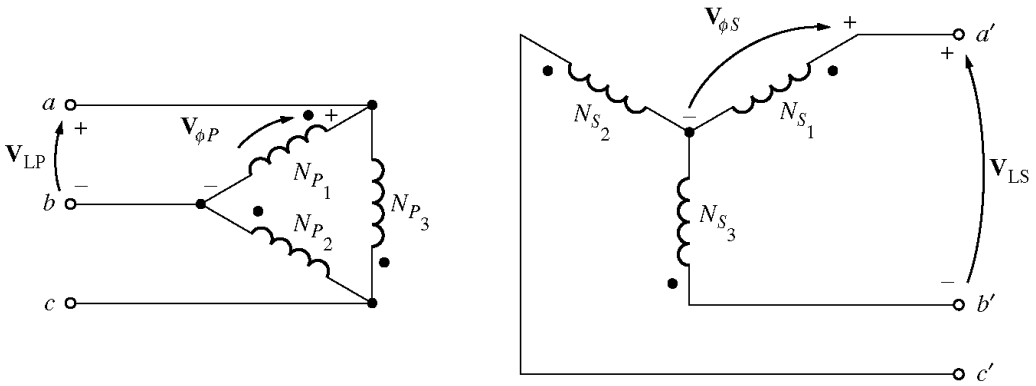
and the voltage regulation is

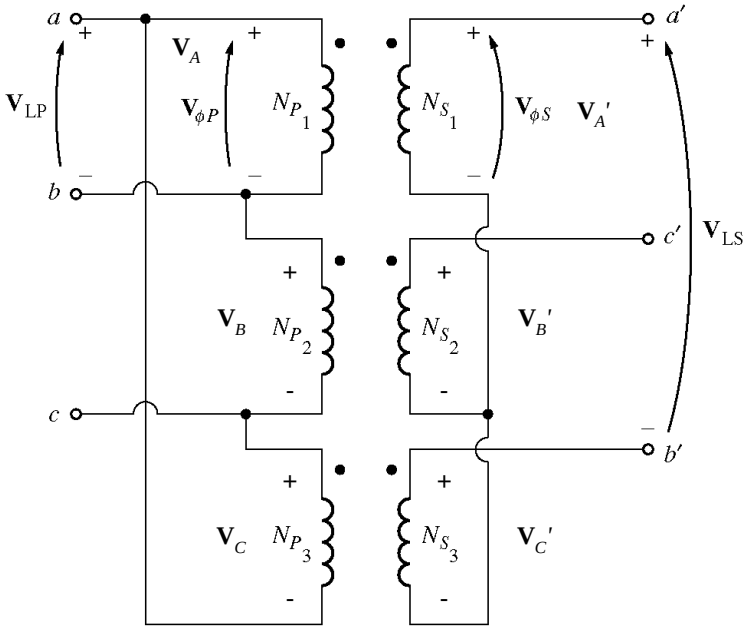
$$\text{VR} = \frac{1.054 - 1.00}{1.00} \times 100\% = 5.4\%$$

(f) The efficiency of the transformer at 50 Hz is almost the same as the efficiency at 60 Hz (just slightly less), but the total apparent power rating of the transformer at 50 Hz must be less than the apparent power rating at 60 Hz by the ratio 50/60. In other words, the efficiencies are similar, but the power handling capability is reduced.

2-22. Prove that the three-phase system of voltages on the secondary of the Δ-Y transformer shown in Figure 2-37*c* lags the three-phase system of voltages on the primary of the transformer by 30°.

SOLUTION The figure is reproduced below:





Assume that the phase voltages on the primary side are given by

$$\mathbf{V}_A = V_{\phi P}\angle 0° \qquad \mathbf{V}_B = V_{\phi P}\angle -120° \qquad \mathbf{V}_C = V_{\phi P}\angle 120°$$

Then the phase voltages on the secondary side are given by

$$\mathbf{V}_A' = V_{\phi S}\angle 0° \qquad \mathbf{V}_B' = V_{\phi S}\angle -120° \qquad \mathbf{V}_C' = V_{\phi S}\angle 120°$$

where $V_{\phi S} = V_{\phi P}/a$. Since this is a Δ-Y transformer bank, the line voltage $\mathbf{V}_{ab}$ on the primary side is just equal to $\mathbf{V}_A = V_{\phi P}\angle 0°$. The line voltage on the secondary side is given by

$$\mathbf{V}_{a'b'} = \mathbf{V}_A - \mathbf{V}_C = V_{\phi P}\angle 0° - V_{\phi P}\angle 120° = \sqrt{3}V_{\phi P}\angle -30°$$

Note that the line voltage on the secondary side lags the line voltage on the primary side by 30ï‚°.

2-23. A single-phase 10-kVA 480/120-V transformer is to be used as an autotransformer tying a 600-V distribution line to a 480-V load. When it is tested as a conventional transformer, the following values are measured on the primary (480-V) side of the transformer:

Open-circuit test (measured on secondary side)	Short-circuit test (measured on primary side)
V_{OC} = 120 V	V_{SC} = 10.0 V
I_{OC} = 1.60 A	I_{SC} = 10.6 A
V_{OC} = 38 W	P_{SC} = 25 W

(a) Find the per-unit equivalent circuit of this transformer when it is connected in the conventional manner. What is the efficiency of the transformer at rated conditions and unity power factor? What is the voltage regulation at those conditions?

(b) Sketch the transformer connections when it is used as a 600/480-V step-down autotransformer.

(c) What is the kilovoltampere rating of this transformer when it is used in the autotransformer connection?

(d) Answer the questions in *(a)* for the autotransformer connection.

SOLUTION

(a) The base impedance of this transformer referred to the primary side is

$$Z_{\text{base},P} = \frac{(V_P)^2}{S} = \frac{(480\text{ V})^2}{10\text{ kVA}} = 23.04\ \Omega$$

The base impedance of this transformer referred to the secondary side is

$$Z_{\text{base},P} = \frac{(V_P)^2}{S} = \frac{(120\text{ V})^2}{10\text{ kVA}} = 1.44\ \Omega$$

The open circuit test yields the values for the excitation branch (referred to the *secondary* side):

$$|Y_{EX}| = \frac{I_{\phi,OC}}{V_{\phi,OC}} = \frac{1.60\text{ A}}{120\text{ V}} = 0.01333\ \text{S}$$

$$\theta = -\cos^{-1}\left(\frac{P_{OC}}{V_{OC}\ I_{OC}}\right) = -\cos^{-1}\left(\frac{38\text{ W}}{(120\text{ V})(1.60\text{ A})}\right) = -78.6°$$

$$Y_{EX} = G_C - jB_M = 0.01333\angle -78.6° = 0.002635 - j0.01307$$

$$R_C = 1/G_C = 380\ \Omega$$

$$X_M = 1/B_M = 76.5\ \Omega$$

The excitation branch elements can be expressed in per-unit as

$$R_C = \frac{380\ \Omega}{1.44\ \Omega} = 263\text{ pu} \qquad X_M = \frac{76.5\ \Omega}{1.44\ \Omega} = 53.1\text{ pu}$$

The short circuit test yields the values for the series impedances (referred to the *primary* side):

$$|Z_{EQ}| = \frac{V_{SC}}{I_{SC}} = \frac{10.0\text{ V}}{10.6\text{ A}} = 0.943\ \Omega$$

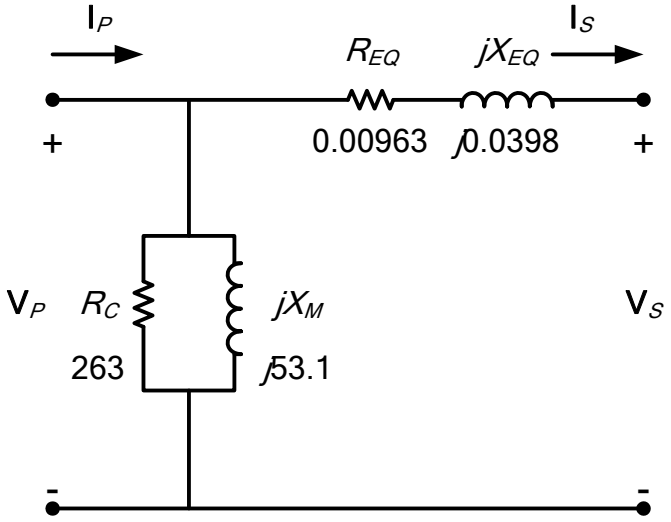
$$\theta = \cos^{-1}\left(\frac{P_{SC}}{V_{SC}\ I_{SC}}\right) = \cos^{-1}\left(\frac{25\text{ W}}{(10.0\text{ V})(10.6\text{ A})}\right) = 76.4°$$

$$Z_{EQ} = R_{EQ} + jX_{EQ} = 0.943\angle 76.4° = 0.222 + j0.917\ \Omega$$

The resulting per-unit impedances are

$$R_{EQ} = \frac{0.222\ \Omega}{23.04\ \Omega} = 0.00963\text{ pu} \qquad X_{EQ} = \frac{0.917\ \Omega}{23.04\ \Omega} = 0.0398\text{ pu}$$

The per-unit equivalent circuit is



At rated conditions and unity power factor, the output power to this transformer would be P_{IN} = 1.0 pu. The input voltage would be

$$\mathbf{V}_P = \mathbf{V}_S + \mathbf{I}_S Z_{\text{EQ}}$$

$$\mathbf{V}_P = 1\angle 0°\text{ V} + (1\angle 0°)(0.00963 + j0.0398\ \Omega) = 1.01\angle 2.23°\text{ pu}$$

The core losses (in resistor R_C) would be

$$P_{\text{core}} = \frac{V^2}{R_C} = \frac{(1.01)^2}{263} = 0.00388\text{ pu}$$

The copper losses (in resistor R_{EQ}) would be

$$P_{\text{CU}} = I^2 R_{\text{EQ}} = (1.0)^2\,(0.00963) = 0.00963\text{ pu}$$

The input power of the transformer would be

$$P_{\text{IN}} = P_{\text{OUT}} - P_{\text{CU}} - P_{\text{core}} = 1.0 + 0.00963 + 0.00388 = 1.0135$$

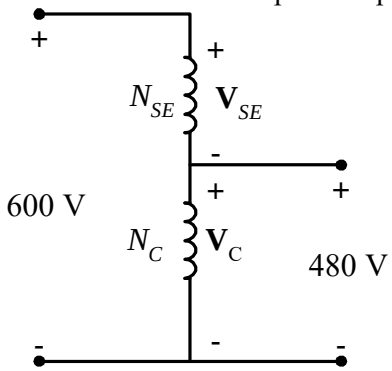
and the transformer efficiency would be

$$\eta = \frac{P_{\text{OUT}}}{P_{\text{IN}}} \times 100\% = \frac{1.0}{1.0135} \times 100\% = 98.7\%$$

The voltage regulation of the transformer is

$$\text{VR} = \frac{1.01 - 1.00}{1.00} \times 100\% = 1.0\%$$

(b) The autotransformer connection for 600/480 V stepdown operation is



(c) When used as an autotransformer, the kVA rating of this transformer becomes:

$$S_{\text{IO}} = \frac{N_C + N_{\text{SE}}}{N_{\text{SE}}} S_W = \frac{4+1}{1}(10 \text{ kVA}) = 50 \text{ kVA}$$

(d) As an autotransformer, the per-unit series impedance Z_{EQ} is decreased by the reciprocal of the power advantage, so the series impedance becomes

$$R_{EQ} = \frac{0.00963}{5} = 0.00193 \text{ pu}$$
$$X_{EQ} = \frac{0.0398}{5} = 0.00796 \text{ pu}$$

while the magnetization branch elements are basically unchanged. At rated conditions and unity power factor, the output power to this transformer would be P_{OUT} = 1.0 pu. The input voltage would be

$$\mathbf{V}_P = \mathbf{V}_S + \mathbf{I}_S Z_{\text{EQ}}$$
$$\mathbf{V}_P = 1\angle 0° \text{ V} + (1\angle 0°)(0.00193 + j0.00796\ \Omega) = 1.002\angle 0.5° \text{ pu}$$

The core losses (in resistor R_C) would be

$$P_{\text{core}} = \frac{V^2}{R_C} = \frac{(1.002)^2}{263} = 0.00382 \text{ pu}$$

The copper losses (in resistor R_{EQ}) would be

$$P_{\text{CU}} = I^2 R_{\text{EQ}} = (1.0)^2 (0.00193) = 0.0019 \text{ pu}$$

The input power of the transformer would be

$$P_{\text{IN}} = P_{\text{OUT}} - P_{\text{CU}} - P_{\text{core}} = 1.0 + 0.0019 + 0.00382 = 1.0057$$

and the transformer efficiency would be

$$\eta = \frac{P_{\text{OUT}}}{P_{\text{IN}}} \times 100\% = \frac{1.0}{1.0057} \times 100\% = 99.4\%$$

The voltage regulation of the transformer is

$$\text{VR} = \frac{1.0057 - 1.00}{1.00} \times 100\% = 0.6\%$$

3-4. A three-phase Y-connected four-pole winding is installed in 24 slots on a stator. There are 40 turns of wire in each slot of the windings. All coils in each phase are connected in series. The flux per pole in the machine is 0.060 Wb, and the speed of rotation of the magnetic field is 1800 r/min.

(a) What is the frequency of the voltage produced in this winding?

(b) What are the resulting phase and terminal voltages of this stator?

SOLUTION

(a) The frequency of the voltage produced in this winding is

$$f_{se} = \frac{n_{sm}P}{120} = \frac{(1800 \text{ r/min})(4 \text{ poles})}{120} = 60 \text{ Hz}$$

(b) There are 24 slots on this stator, with 40 turns of wire per slot. Since this is a four-pole machine, there are two sets of coils (in 8 slots) associated with each phase. The voltage in the coils in *one pair* of slots is

$$E_A = \sqrt{2}\pi N_C \phi f = \sqrt{2}\pi(40 \text{ t})(0.060 \text{ Wb})(60 \text{ Hz}) = 640 \text{ V}$$

There are eight slots associated with each phase, and the all of the coils in a slot are connected in series, so the total phase voltage is

$$V_\phi = 8(640 \text{ V}) = 5120 \text{ V}$$

Since the machine is Y-connected, $V_L = \sqrt{3}V_\phi = 8868 \text{ V}$.

3-5. A three-phase Δ-connected six-pole winding is installed in 36 slots on a stator. There are 150 turns of wire in each slot of the windings. All coils in each phase are connected in series. The flux per pole in the machine is 0.060 Wb, and the speed of rotation of the magnetic field is 1000 r/min.

(a) What is the frequency of the voltage produced in this winding?

(b) What are the resulting phase and terminal voltages of this stator?

SOLUTION

(a) The frequency of the voltage produced in this winding is

$$f_{se} = \frac{n_{sm}P}{120} = \frac{(1000 \text{ r/min})(6 \text{ poles})}{120} = 50 \text{ Hz}$$

(b) There are 36 slots on this stator, with 150 turns of wire per slot. Since this is a six-pole machine, there are three sets of coils (in 12 slots) associated with each phase. The voltage in the coils in *one pair* of slots is

$$E_A = \sqrt{2}\pi N_C \phi f = \sqrt{2}\pi(150 \text{ t})(0.060 \text{ Wb})(50 \text{ Hz}) = 2000 \text{ V}$$

There are 12 slots associated with each phase, and the all of the coils in a slot are connected in series, so the total phase voltage is

$$V_\phi = 12(2000 \text{ V}) = 24{,}000 \text{ V}$$

Since the machine is Δ -connected, $V_L = V_\phi = 24{,}000 \text{ V}$.

3-6. A three-phase Y-connected 60-Hz two-pole synchronous machine has a stator with 5000 turns of wire per phase. What rotor flux would be required to produce a terminal (line-to-line) voltage of 13.2 kV?

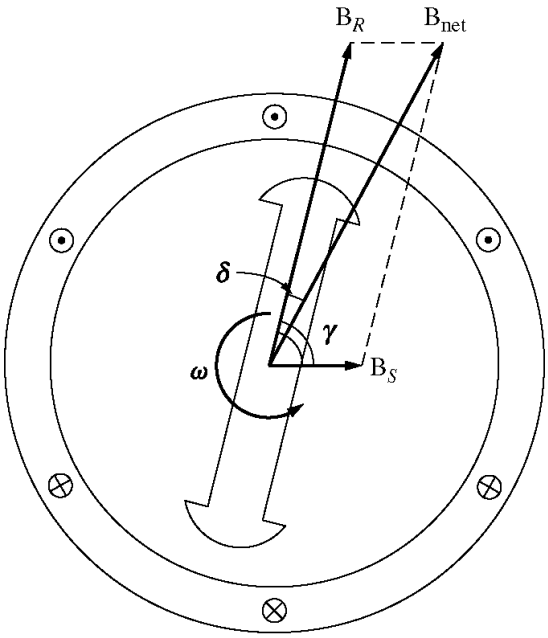
SOLUTION The phase voltage of this machine should be $V_\phi = V_L / \sqrt{3} = 7621$ V. The induced voltage per phase in this machine (which is equal to V_ϕ at no-load conditions) is given by the equation

$$E_A = \sqrt{2}\pi N_C \phi f$$

so

$$\phi = \frac{E_A}{\sqrt{2}\pi N_C f} = \frac{7621 \text{ V}}{\sqrt{2}\pi (5000 \text{ t})(60 \text{ Hz})} = 0.0057 \text{ Wb}$$

3-8. If an ac machine has the rotor and stator magnetic fields shown in Figure P3-1, what is the direction of the induced torque in the machine? Is the machine acting as a motor or generator?



SOLUTION Since $\boldsymbol{\tau}_{ind} = k \times \mathbf{B}_R \mathbf{B}_{net}$, the induced torque is *clockwise*, opposite the direction of motion. The machine is acting as a generator.

4-3. Assume that the field current of the generator in Problem 4-2 has been adjusted to a value of 5 A.

(a) What will the terminal voltage of this generator be if it is connected to a Δ-connected load with an impedance of $24\angle 25°\ \Omega$?

(b) Sketch the phasor diagram of this generator.

(c) What is the efficiency of the generator at these conditions?

(d) Now assume that another identical Δ-connected load is to be paralleled with the first one. What happens to the phasor diagram for the generator?

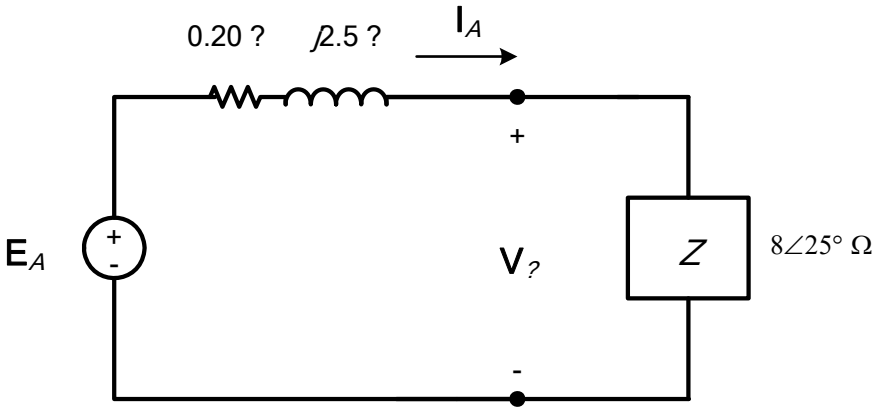
(e) What is the new terminal voltage after the load has been added?

(f) What must be done to restore the terminal voltage to its original value?

SOLUTION

(a) If the field current is 5.0 A, the open-circuit terminal voltage will be about 16,500 V, and the open-circuit phase voltage in the generator (and hence E_A) will be $16{,}500/\sqrt{3} = 9526$ V.

The load is Δ-connected with three impedances of $24\angle 25°\ \Omega$. From the Y-Δ transform, this load is equivalent to a Y-connected load with three impedances of $8\angle 25°\ \Omega$. The resulting per-phase equivalent circuit is shown below:



The magnitude of the phase current flowing in this generator is

$$I_A = \frac{E_A}{|R_A + jX_S + Z|} = \frac{9526\text{ V}}{|0.2 + j2.5 + 8\angle 25°|} = \frac{9526\text{ V}}{9.49\ \Omega} = 1004\text{ A}$$

Therefore, the magnitude of the phase voltage is

$$V_\phi = I_A Z = (1004\text{ A})(8\ \Omega) = 8032\text{ V}$$

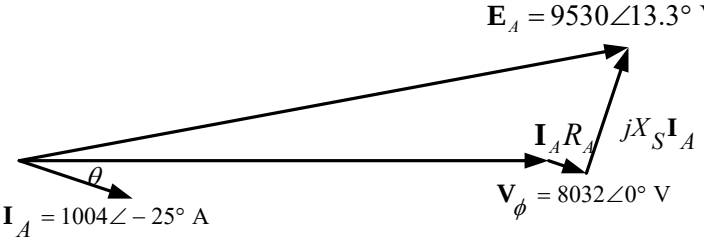
and the terminal voltage is

$$V_T = \sqrt{3}\ V_\phi = \sqrt{3}\ (8032\text{ V}) = 13{,}910\text{ V}$$

(b) Armature current is $\mathbf{I}_A = 1004\angle -25°$ A, and the phase voltage is $\mathbf{V}_\phi = 8032\angle 0°$ V. Therefore, the internal generated voltage is

$$\mathbf{E}_A = \mathbf{V}_\phi + R_A\mathbf{I}_A + jX_S\mathbf{I}_A$$
$$\mathbf{E}_A = 8032\angle 0° + (0.20\ \Omega)(1004\angle -25°\text{ A}) + j(2.5\ \Omega)(1004\angle -25°\text{ A})$$
$$\mathbf{E}_A = 9530\angle 13.3°\text{ V}$$

The resulting phasor diagram is shown below (not to scale):



(c) The efficiency of the generator under these conditions can be found as follows:

$$P_{\text{OUT}} = 3\ V_\phi\ I_A\ \cos\ \theta = 3(8032\ \text{V})(1004\ \text{A})\cos(-25°) = 21.9\ \text{MW}$$

$$P_{\text{CU}} = 3I_A{}^2 R_A = 3(1004\ \text{A})^2\ (0.2\ \Omega) = 605\ \text{kW}$$

$$P_{\text{F\&W}} = 1\ \text{MW}$$

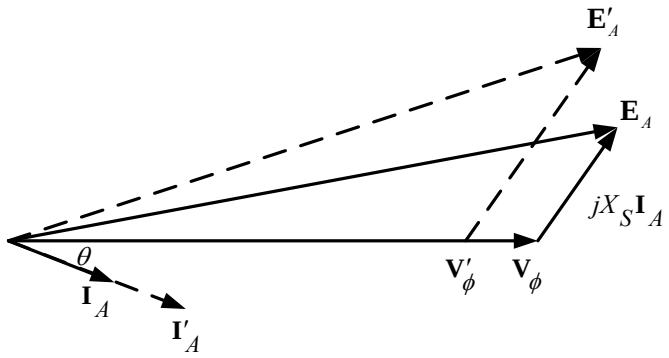
$$P_{\text{core}} = 1.5\ \text{MW}$$

$$P_{\text{stray}} = (\text{assumed } 0)$$

$$P_{\text{IN}} = P_{\text{OUT}} + P_{\text{CU}} + P_{\text{F\&W}} + P_{\text{core}} + P_{\text{stray}} = 25\ \text{MW}$$

$$\eta = \frac{P_{\text{OUT}}}{P_{\text{IN}}} \times 100\% = \frac{21.9\ \text{MW}}{25\ \text{MW}} \times 100\% = 87.6\%$$

(d) To get the basic idea of what happens, we will ignore the armature resistance for the moment. If the field current and the rotational speed of the generator are constant, then the magnitude of $\mathbf{E}_A\ (= K\phi\omega)$ is constant. The quantity $jX_S\mathbf{I}_A$ increases in length at the same angle, while the magnitude of $\mathbf{E}_A$ must remain constant. Therefore, $\mathbf{E}_A$ "swings" out along the arc of constant magnitude until the new $jX_S\mathbf{I}_S$ fits exactly between $\mathbf{V}_\phi$ and $\mathbf{E}_A$.



(e) The new impedance per phase will be half of the old value, so $Z = 4\angle 25°\ \Omega$. The magnitude of the phase current flowing in this generator is

$$I_A = \frac{E_A}{|R_A + jX_S + Z|} = \frac{9526\ \text{V}}{|0.2 + j2.5 + 4\angle 25°|} = \frac{9526\ \text{V}}{5.67\ \Omega} = 1680\ \text{A}$$

Therefore, the magnitude of the phase voltage is

$$V_\phi = I_A Z = (1680\ \text{A})(4\ \Omega) = 6720\ \text{V}$$

and the terminal voltage is

$$V_T = \sqrt{3}\ V_\phi = \sqrt{3}\ (6720\ \text{V}) = 11{,}640\ \text{V}$$

(f) To restore the terminal voltage to its original value, increase the field current I_F.

4-8. A 200-MVA, 12-kV, 0.85-PF-lagging, 50-Hz, 20-pole, Y-connected water turbine generator has a per-unit synchronous reactance of 0.9 and a per-unit armature resistance of 0.1. This generator is operating in parallel with a large power system (infinite bus).

(a) What is the speed of rotation of this generator's shaft?

(b) What is the magnitude of the internal generated voltage E_A at rated conditions?

(c) What is the torque angle of the generator at rated conditions?

(d) What are the values of the generator's synchronous reactance and armature resistance in ohms?

(e) If the field current is held constant, what is the maximum power possible out of this generator? How much reserve power or torque does this generator have at full load?

(f) At the absolute maximum power possible, how much reactive power will this generator be supplying or consuming? Sketch the corresponding phasor diagram. (Assume I_F is still unchanged.)

SOLUTION

(a) The speed of rotation of this generator's shaft is

$$n_{\text{sync}} = \frac{120 f_{se}}{P} = \frac{120(50\text{ Hz})}{20} = 300\text{ r/min}$$

(b) The per-unit phase voltage at rated conditions is $\mathbf{V}_\phi = 1.0\angle 0°$ and the per-unit phase current at rated conditions is $\mathbf{I}_A = 1.0\angle -25.8°$ (since the power factor is 0.9 lagging), so the per-unit internal generated voltage is

$$\mathbf{E}_A = \mathbf{V}_\phi + R_A\mathbf{I}_A + jX_S\mathbf{I}_A$$
$$\mathbf{E}_A = 1\angle 0° + (0.1)(1\angle -25.8°) + j(0.9)(1\angle -25.8°)$$
$$\mathbf{E}_A = 1.69\angle 27.4°\text{ pu}$$

The base phase voltage is

$$V_{\phi,\text{base}} = 12\text{ kV}/\sqrt{3} = 6928\text{ V}$$

so the internal generated voltage is

$$\mathbf{E}_A = (1.69\angle 27.4°\text{ pu})(6928\text{ V}) = 11{,}710\angle 27.4°\text{ V}$$

(c) The torque angle of the generator is $\delta = 27.4°$.

(d) The base impedance of the generator is

$$Z_{\text{base}} = \frac{3\,V_{\phi,\text{base}}{}^2}{S_{\text{base}}} = \frac{3(6928\text{ V})^2}{200{,}000{,}000\text{ VA}} = 0.72\ \Omega$$

Therefore the synchronous reactance is

$$X_S = (0.9)(0.72\ \Omega) = 0.648\ \Omega$$

and the armature resistance is

$$R_A = (0.1)(0.72\ \Omega) = 0.072\ \Omega$$

(e) If the field current is held constant (and the armature resistance is ignored), the power out of this generator is given by

$$P = \frac{3V_\phi E_A}{X_S}\sin\delta$$

The max power is given by

$$P_{\max} = \frac{3V_\phi E_A}{X_S}\sin 90° = \frac{3(6928\text{ V})(11{,}710\text{ V})}{0.648\ \Omega} = 376\text{ MW}$$

Since the full load power is $P = (200\text{ MVA})(0.85) = 170\text{ MW}$, this generator is supplying 45% of the maximum possible power at full load conditions.

(f) At the maximum power possible, the torque angle $\delta = 90°$, so the phasor $\mathbf{E}_A$ will be at an angle of 90°, and the current flowing will be

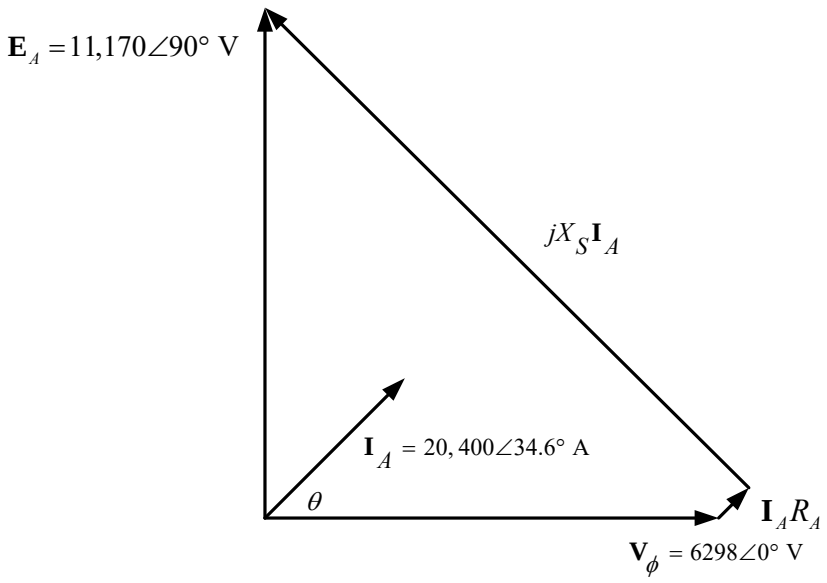
$$\mathbf{E}_A = \mathbf{V}_\phi + R_A\mathbf{I}_A + jX_S\mathbf{I}_A$$

$$\mathbf{I}_A = \frac{\mathbf{E}_A - \mathbf{V}_\phi}{R_A + jX_S}$$

$$\mathbf{I}_A = \frac{11{,}710\angle 90°\text{ kV} - 6298\angle 0°\text{ kV}}{0.072 + j0.648\ \Omega} = 20{,}400\angle 34.6°\text{ A}$$

The impedance angle $\theta = -34.6°$, and the reactive power supplied by the generator is

$$Q = 3V_\phi I_A \sin\theta = 3(6298\text{ V})(20{,}400\text{ A})\sin(-34.6°) = -219\text{ Mvar}$$



4-13. A 25-MVA, 12.2-kV, 0.9-PF-lagging, three-phase, two-pole, Y-connected, 60-Hz synchronous generator was tested by the open-circuit test, and its air-gap voltage was extrapolated with the following results:

Open-circuit test

Field current, A	275	320	365	380	475	570
Line voltage, kV	12.2	13.0	13.8	14.1	15.2	16.0
Extrapolated air-gap voltage, kV	13.3	15.4	17.5	18.3	22.8	27.4

The short-circuit test was then performed with the following results:

Short-circuit test

Field current, A	275	320	365	380	475	570
Armature current, A	890	1040	1190	1240	1550	1885

The armature resistance is 0.6 Ω per phase.

(a) Find the unsaturated synchronous reactance of this generator in ohms per phase and in per-unit.

(b) Find the approximate saturated synchronous reactance X_S at a field current of 380 A. Express the answer both in ohms per phase and in per-unit.

(c) Find the approximate saturated synchronous reactance at a field current of 475 A. Express the answer both in ohms per phase and in per-unit.

(d) Find the short-circuit ratio for this generator.

(e) What is the internal generated voltage of this generator at rated conditions?

(f) What field current is required to achieve rated voltage at rated load?

SOLUTION

(a) The unsaturated synchronous reactance of this generator is the same at any field current, so we will look at it at a field current of 380 A. The extrapolated air-gap voltage at this point is 18.3 kV, and the short-circuit current is 1240 A. Since this generator is Y-connected, the phase voltage is $V_\phi = 18.3 \text{ kV}/\sqrt{3} = 10{,}566 \text{ V}$ and the armature current is $I_A = 1240 \text{ A}$. Therefore, the *unsaturated* synchronous reactance is

$$X_{Su} = \frac{10{,}566 \text{ V}}{1240 \text{ A}} = 8.52 \ \Omega$$

The base impedance of this generator is

$$Z_{\text{base}} = \frac{3\ V_{\phi,\text{base}}{}^2}{S_{\text{base}}} = \frac{3(7044 \text{ V})^2}{25{,}000{,}000 \text{ VA}} = 5.95 \ \Omega$$

Therefore, the per-unit unsaturated synchronous reactance is

$$X_{Su,\text{pu}} = \frac{8.52 \ \Omega}{5.95 \ \Omega} = 1.43$$

(b) The saturated synchronous reactance at a field current of 380 A can be found from the OCC and the SCC. The OCC voltage at I_F = 380 A is 14.1 kV, and the short-circuit current is 1240 A. Since this generator is Y-connected, the corresponding phase voltage is $V_\phi = 14.1 \text{ kV}/\sqrt{3} = 8141 \text{ V}$ and the armature current is $I_A = 1240 \text{ A}$. Therefore, the *saturated* synchronous reactance is

$$X_S = \frac{8141 \text{ V}}{1240 \text{ A}} = 6.57 \ \Omega$$

and the per-unit unsaturated synchronous reactance is

$$X_{S,\text{pu}} = \frac{8.52 \ \Omega}{5.95 \ \Omega} = 1.10$$

(c) The saturated synchronous reactance at a field current of 475 A can be found from the OCC and the SCC. The OCC voltage at I_F = 475 A is 15.2 kV, and the short-circuit current is 1550 A. Since this generator is Y-connected, the corresponding phase voltage is $V_\phi = 15.2 \text{ kV}/\sqrt{3} = 8776 \text{ V}$ and the armature current is $I_A = 1550 \text{ A}$. Therefore, the *saturated* synchronous reactance is

$$X_S = \frac{8776 \text{ V}}{1550 \text{ A}} = 5.66 \ \Omega$$

and the per-unit unsaturated synchronous reactance is

$$X_{S,\text{pu}} = \frac{5.66 \ \Omega}{5.95 \ \Omega} = 0.951$$

(d) The rated voltage of this generator is 12.2 kV, which requires a field current of 275 A. The rated line and armature current of this generator is

$$I_L = \frac{25 \text{ MVA}}{\sqrt{3}(12.2 \text{ kV})} = 1183 \text{ A}$$

The field current required to produce a short-circuit current of 1183 A is about 365 A. Therefore, the short-circuit ratio of this generator is

$$\text{SCR} = \frac{275 \text{ A}}{365 \text{ A}} = 0.75$$

(e) The internal generated voltage of this generator at rated conditions would be calculated using the saturated synchronous reactance, which is about $X_S = 6.57 \ \Omega$ if the field current is 380 A. Since the power factor is 0.9 lagging, the armature current is $\mathbf{I}_A = 1183 \angle -25.8° \text{ A}$. The phase voltage is $V_\phi = 12{,}200/\sqrt{3} \angle 0° \text{ V} = 7044 \angle 0° \text{ V}$. Therefore,

$$\mathbf{E}_A = \mathbf{V}_\phi + R_A \mathbf{I}_A + jX_S \mathbf{I}_A$$
$$\mathbf{E}_A = 7044 \angle 0° + (0.60 \ \Omega)(1183 \angle -25.8° \text{ A}) + j(6.57 \ \Omega)(1183 \angle -25.8° \text{ A})$$
$$\mathbf{E}_A = 12{,}930 \angle 31.2° \text{ V}$$

(f) If the internal generated voltage is 12,930 V per phase, the corresponding line value would be $(12{,}930 \text{ V})(\sqrt{3}) = 22{,}400 \text{ V}$. This would require a field current of about 470 A.

4-15. A three-phase Y-connected synchronous generator is rated 120 MVA, 13.8 kV, 0.8-PF-lagging, and 60 Hz. Its synchronous reactance is 1.2 Ω per phase, and its armature resistance is 0.1 Ω per phase.

(a) What is its voltage regulation?

(b) What would the voltage and apparent power rating of this generator be if it were operated at 50 Hz with the same armature and field losses as it had at 60 Hz?

(c) What would the voltage regulation of the generator be at 50 Hz?

SOLUTION

(a) The rated armature current is

$$I_A = I_L = \frac{S}{\sqrt{3}\ V_T} = \frac{120\text{ MVA}}{\sqrt{3}(13.8\text{ kV})} = 5020\text{ A}$$

The power factor is 0.8 lagging, so $\mathbf{I}_A = 5020\angle -36.87°\text{ A}$. The phase voltage is 13.8 kV / $\sqrt{3}$ = 7967 V. Therefore, the internal generated voltage is

$$\mathbf{E}_A = \mathbf{V}_\phi + R_A\mathbf{I}_A + jX_S\mathbf{I}_A$$
$$\mathbf{E}_A = 7967\angle 0° + (0.1\ \Omega)(5020\angle -36.87°\text{ A}) + j(1.2\ \Omega)(5020\angle -36.87°\text{ A})$$
$$\mathbf{E}_A = 12{,}800\angle 20.7°\text{ V}$$

The resulting voltage regulation is

$$\text{VR} = \frac{12{,}800 - 7967}{7967} \times 100\% = 60.7\%$$

(b) If the generator is to be operated at 50 Hz with the same armature and field losses as at 60 Hz (so that the windings do not overheat), then its armature and field currents must not change. Since the voltage of the generator is directly proportional to the speed of the generator, the voltage rating (and hence the apparent power rating) of the generator will be reduced by a factor of 5/6.

$$V_{T,\text{rated}} = \frac{5}{6}(13.8\text{ kV}) = 11.5\text{ kV}$$

$$S_{\text{rated}} = \frac{5}{6}(120\text{ MVA}) = 100\text{ MVA}$$

Also, the synchronous reactance will be reduced by a factor of 5/6.

$$X_S = \frac{5}{6}(1.2\ \Omega) = 1.00\ \Omega$$

(c) At 50 Hz rated conditions, the armature current would be

$$I_A = I_L = \frac{S}{\sqrt{3}\ V_T} = \frac{100\text{ MVA}}{\sqrt{3}(11.5\text{ kV})} = 5020\text{ A}$$

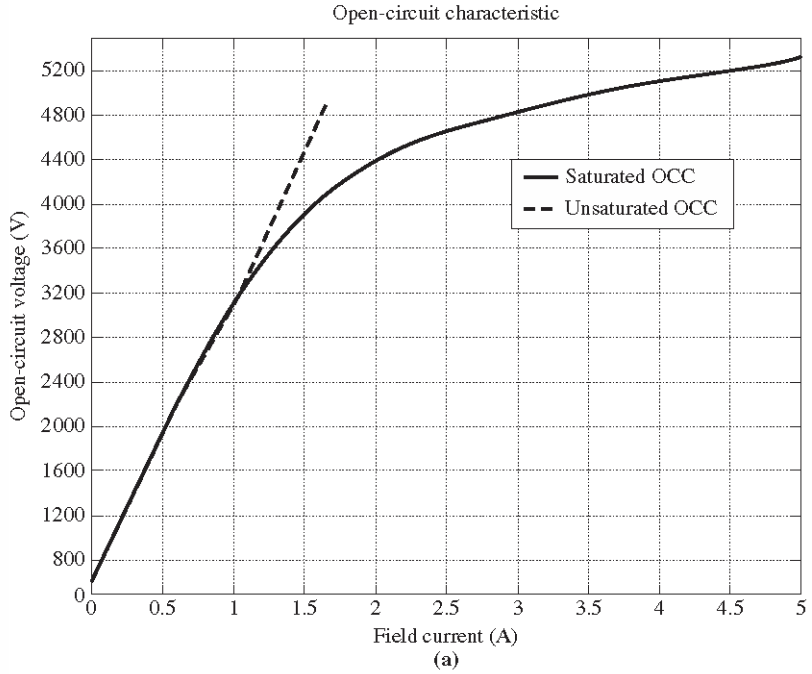
The power factor is 0.8 lagging, so $\mathbf{I}_A = 5020\angle -36.87°\text{ A}$. The phase voltage is 11.5 kV / $\sqrt{3}$ = 6640 V. Therefore, the internal generated voltage is

$$\mathbf{E}_A = \mathbf{V}_\phi + R_A\mathbf{I}_A + jX_S\mathbf{I}_A$$
$$\mathbf{E}_A = 6640\angle 0° + (0.1\ \Omega)(5020\angle -36.87°\text{ A}) + j(1.0\ \Omega)(5020\angle -36.87°\text{ A})$$
$$\mathbf{E}_A = 10{,}300\angle 18.8°\text{ V}$$

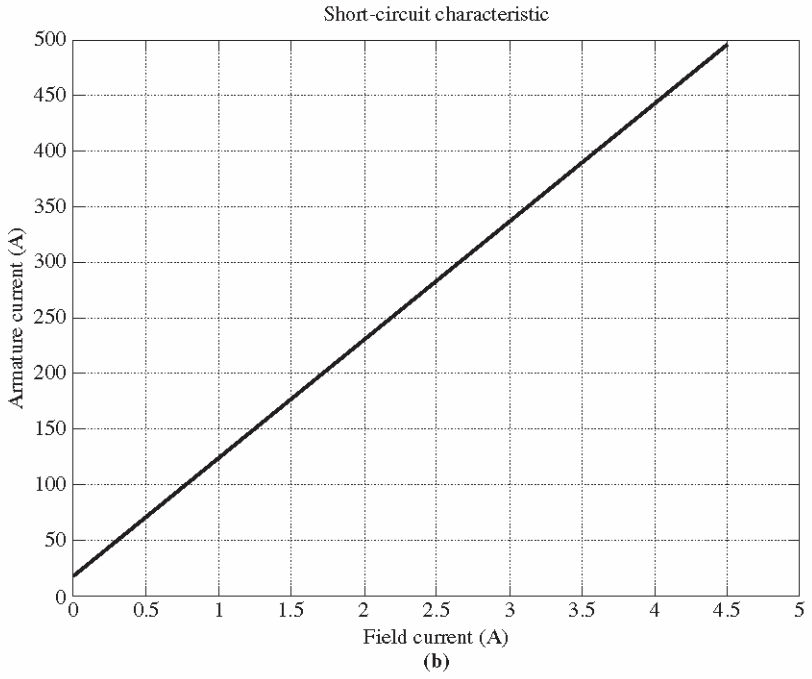
The resulting voltage regulation is

$$\text{VR} = \frac{10{,}300 - 6640}{6640} \times 100\% = 55.1\%$$

Problems 4-16 to 4-26 refer to a six-pole Y-connected synchronous generator rated at **500 kVA**, 3.2 kV, 0.9 PF lagging, and 60 Hz. Its armature resistance R_A is 0.7 Ω. The core losses of this generator at rated conditions are 8 kW, and the friction and windage losses are 10 kW. The open-circuit and short-circuit characteristics are shown in Figure P4-2



(a)



(b)

FIGURE P4–2
(a) Open-circuit characteristic curve for the generator in Problems 4–16 to 4–26. *(b)* Short-circuit characteristic curve for the generator in Problems 4–16 to 4–26.

4-17. *(a)* What are the rated current and internal generated voltage of this generator? *(b)* What field current does this generator require to operate at the rated voltage, current, and power factor?

SOLUTION

(a) The rated line and armature current for this generator is

$$I_A = I_L = \frac{S}{\sqrt{3}\ V_T} = \frac{500\text{ kVA}}{\sqrt{3}(3200\text{ V})} = 90.2\text{ A}$$

The power factor is 0.9 lagging, so $\mathbf{I}_A = 90.2\angle -25.8°\text{ A}$. The rated phase voltage is $V_\phi = 3.2\text{ kV} / \sqrt{3} = 1850\text{ V}$. The saturated synchronous reactance at rated conditions was found to be 15.7 Ω in the previous problem. Therefore, the internal generated voltage is

$$\mathbf{E}_A = \mathbf{V}_\phi + R_A\mathbf{I}_A + jX_S\mathbf{I}_A$$
$$\mathbf{E}_A = 1850\angle 0° + (0.7\ \Omega)(90.2\angle -25.8°\text{ A}) + j(15.7\ \Omega)(90.2\angle -25.8°\text{ A})$$
$$\mathbf{E}_A = 2814\angle 26.3°\text{ V}$$

(b) This internal generated voltage corresponds to a no-load terminal voltage of $\sqrt{3}(2814) = 4874\text{ V}$. From the open-circuit characteristic, the required field current would be 3.2 A.

4-20. What are the electrical losses in this generator at rated conditions?

SOLUTION The current flow at rated conditions is $I_A = 90.2\text{ A}$, so the electrical losses are

$$P_{\text{CU}} = 3I_A^2 R_A = 3(90.2\text{ A})^2(0.7\ \Omega) = 17.1\text{ kW}$$

4-26. Assume that the generator is connected to a 3200-V infinite bus, and that its field current has been adjusted so that it is supplying rated power and power factor to the bus. You may ignore the armature resistance R_A when answering the following questions.

(a) What would happen to the real and reactive power supplied by this generator if the field flux (and therefore E_A) is reduced by 5 percent.

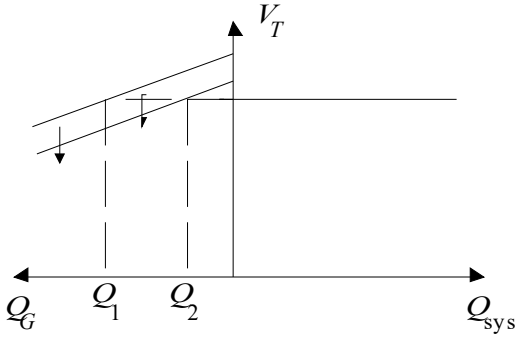
(b) Plot the real power supplied by this generator as a function of the flux ϕ as the flux is varied from 80% to 100% of the flux at rated conditions.

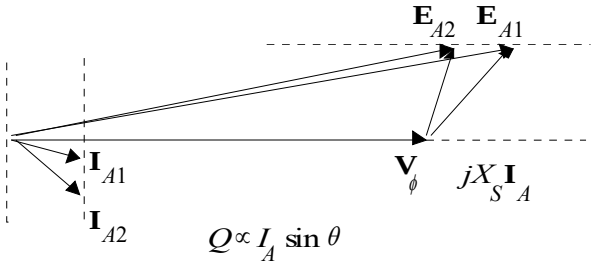
(c) Plot the reactive power supplied by this generator as a function of the flux ϕ as the flux is varied from 80% to 100% of the flux at rated conditions.

(d) Plot the line current supplied by this generator as a function of the flux ϕ as the flux is varied from 80% to 100% of the flux at rated conditions.

SOLUTION

(a) If the field flux is reduced by 5%, nothing would happen to the real power. The reactive power supplied would decrease as shown below.





(c) If armature resistance is ignored, the internal generated voltage $\mathbf{E}_A$ will increase as flux increases, but the quantity $E_A \sin\delta$ will remain constant. Therefore, the voltage for any flux can be found from the expression

$$E_A = \left(\frac{\phi}{\phi_r}\right) E_{Ar}$$

and the angle δ for any E_A can be found from the expression

$$\delta = \sin^{-1}\left(\frac{E_{Ar}}{E_A}\sin\delta_r\right)$$

where ϕ is the flux in the machine, ϕ_r is the flux at rated conditions, E_{Ar} is the magnitude of the internal generated voltage at rated conditions, and δ_r is the angle of the internal generated voltage at rated conditions. From this information, we can calculate $\mathbf{I}_A$ for any given load from equation

$$\mathbf{I}_A = \frac{\mathbf{E}_A - \mathbf{V}_\phi}{jX_S}$$

and the resulting reactive power from the equation

$$Q = 3V_\phi I_A \sin\theta$$

where θ is the impedance angle, which is the negative of the current angle.

The rated line and armature current for this generator is

$$I_A = I_L = \frac{S}{\sqrt{3}\ V_T} = \frac{500\text{ kVA}}{\sqrt{3}(3200\text{ V})} = 90.2\text{ A}$$

The power factor is 0.9 lagging, so $\mathbf{I}_A = 90.2\angle -25.8°$ A. The rated phase voltage is V_ϕ = 3.2 kV / $\sqrt{3}$ = 1850 V. The saturated synchronous reactance at rated conditions was found to be 15.7 Ω in the previous problem. Therefore, the internal generated voltage (ignoring R_A) is

$$\mathbf{E}_A = \mathbf{V}_\phi + jX_S\mathbf{I}_A$$
$$\mathbf{E}_A = 1850\angle 0° + j(15.7\ \Omega)(90.2\angle -25.8°\ \text{A})$$
$$\mathbf{E}_A = 2776\angle 27.3°\ \text{V}$$

so $E_{Ar} = 2776$ V and $\delta_r = 27.3°$. A MATLAB program that calculates the reactive power supplied

4-27. Two identical 2.5-MVA, 1200-V, 0.8-PF-lagging, 60 Hz, three-phase synchronous generators are connected in parallel to supply a load. The prime movers of the two generators happen to have different speed droop characteristics. When the field currents of the two generators are equal, one delivers 1200 A at 0.9 PF lagging, while the other delivers 900 A at 0.75 PF lagging.

(a) What are the real power and the reactive power supplied by each generator to the load?

(b) What is the overall power factor of the load?

(c) In what direction must the field current on each generator be adjusted in order for them to operate at the same power factor?

SOLUTION

(a) The real and reactive powers are

$$P_1 = \sqrt{3}\ V_T\ I_L\ \cos\theta = \sqrt{3}(1200\text{ V})(1200\text{ A})(0.9) = 2245\text{ kW}$$

$$Q_1 = \sqrt{3}\ V_T\ I_L\ \sin\theta = \sqrt{3}(1200\text{ V})(1200\text{ A})\sin\left[\cos^{-1}(0.9)\right] = 1087\text{ kVAR}$$

$$P_2 = \sqrt{3}\ V_T\ I_L\ \cos\theta = \sqrt{3}(1200\text{ V})(900\text{ A})(0.75) = 1403\text{ kW}$$

$$Q_2 = \sqrt{3}\ V_T\ I_L\ \sin\theta = \sqrt{3}(1200\text{ V})(900\text{ A})\sin\left[\cos^{-1}(0.75)\right] = 1237\text{ kVAR}$$

(b) The overall power factor can be found from the total real and reactive power supplied to the load.

$$P_{\text{TOT}} = P_1 + P_2 = 2245\text{ kW} + 1403\text{ kW} = 3648\text{ kW}$$

$$Q_{\text{TOT}} = Q_1 + Q_2 = 1087\text{ kVAR} + 1237\text{ kVAR} = 2324\text{ kVAR}$$

The overall power factor is

$$\text{PF} = \cos\left[\tan^{-1}\frac{Q_{\text{TOT}}}{P_{\text{TOT}}}\right] = 0.843\text{ lagging}$$

(c) The field current of generator 1 should be increased, and the field current of generator 2 should be simultaneously decreased.

4-28. A generating station for a power system consists of four 300-MVA, 15-kV, 0.85-PF-lagging synchronous generators with identical speed droop characteristics operating in parallel. The governors on the generators' prime movers are adjusted to produce a 3-Hz drop from no load to full load. Three of these generators are each supplying a steady 200 MW at a frequency of 60 Hz, while the fourth generator (called the *swing generator*) handles all incremental load changes on the system while maintaining the system's frequency at 60 Hz.

(a) At a given instant, the total system loads are 650 MW at a frequency of 60 Hz. What are the no-load frequencies of each of the system's generators?

(b) If the system load rises to 725 MW and the generator's governor set points do not change, what will the new system frequency be?

(c) To what frequency must the no-load frequency of the swing generator be adjusted in order to restore the system frequency to 60 Hz?

(d) If the system is operating at the conditions described in part *(c)*, what would happen if the swing generator were tripped off the line (disconnected from the power line)?

SOLUTION

(a) The full-load power of these generators is $(300\text{ MVA})(0.85) = 255\text{ MW}$ and the droop from no-load to full-load is 3 Hz. Therefore, the slope of the power-frequency curve for these four generators is

$$s_P = \frac{255\text{ MW}}{3\text{ Hz}} = 85\text{ MW/Hz}$$

If generators 1, 2, and 3 are supplying 200 MW each, then generator 4 must be supplying 50 MW. The no-load frequency of the first three generators is

$$P_1 = s_{P1}\left(f_{\text{nl1}} - f_{\text{sys}}\right)$$

$$200\text{ MW} = (85\text{ MW/Hz})\left(f_{\text{nl1}} - 60\text{ Hz}\right)$$

$$f_{\text{nl1}} = 62.35\text{ Hz}$$

The no-load frequency of the fourth generator is

$$P_4 = s_{P4}\left(f_{\text{nl4}} - f_{\text{sys}}\right)$$

$$50\text{ MW} = (85\text{ MW/Hz})\left(f_{\text{nl1}} - 60\text{ Hz}\right)$$

$$f_{\text{nl1}} = 60.59\text{ Hz}$$

(b) The setpoints of generators 1, 2, 3, and 4 do not change, so the new system frequency will be

$$P_{\text{LOAD}} = s_{P1}\left(f_{\text{nl1}} - f_{\text{sys}}\right) + s_{P2}\left(f_{\text{nl2}} - f_{\text{sys}}\right) + s_{P3}\left(f_{\text{nl3}} - f_{\text{sys}}\right) + s_{P4}\left(f_{\text{nl4}} - f_{\text{sys}}\right)$$

$$725\text{ MW} = (85)\left(62.35 - f_{\text{sys}}\right) + (85)\left(62.35 - f_{\text{sys}}\right) + (85)\left(62.35 - f_{\text{sys}}\right) + (85)\left(60.59 - f_{\text{sys}}\right)$$

$$725 = 21049.40 - 340 f_{\text{sys}}$$

$$-20324.40 = -340 f_{\text{sys}}$$

$$f_{\text{sys}} = 59.78\text{ Hz}$$

(c) The governor setpoints of the swing generator must be increased until the system frequency rises back to 60 Hz. At 60 Hz, the other three generators will be supplying 200 MW each, so the swing generator must supply 725 MW – 3(200 MW) = 125 MW at 60 Hz. Therefore, the swing generator's setpoints must be set to

$$P_4 = s_{P4}\left(f_{\text{nl4}} - f_{\text{sys}}\right)$$

$$125\text{ MW} = (85\text{ MW/Hz})\left(f_{\text{nl1}} - 60\text{ Hz}\right)$$

$$f_{\text{nl1}} = 61.47\text{ Hz}$$

(d) If the swing generator trips off the line, the other three generators would have to supply all 725 MW of the load. Therefore, the system frequency will become

$$P_{\text{LOAD}} = s_{P1}\left(f_{\text{nl1}} - f_{\text{sys}}\right) + s_{P2}\left(f_{\text{nl2}} - f_{\text{sys}}\right) + s_{P3}\left(f_{\text{nl3}} - f_{\text{sys}}\right)$$

$$725\text{ MW} = (85)\left(62.35 - f_{\text{sys}}\right) + (85)\left(62.35 - f_{\text{sys}}\right) + (85)\left(62.35 - f_{\text{sys}}\right)$$

$$725 = 15899.25 - 255 f_{\text{sys}}$$

$$-15174.25 = -255 f_{\text{sys}}$$

f_{sys} ⎫59.51 Hz Each generator will supply 241.7 MW to the loads.

4-29. A 100-MVA, 14.4-kV 0.8-PF-lagging, Y-connected synchronous generator has a negligible armature resistance and a synchronous reactance of 1.0 per-unit. The generator is connected in parallel with a 60-Hz, 14.4-kV infinite bus that is capable of supplying or consuming any amount of real or reactive power with no change in frequency or terminal voltage.

(a) What is the synchronous reactance of the generator in ohms?

(b) Repeat part *(d)* for 10, 15, 20, and 25 percent reductions in $\mathbf{E}_A$.

(f) Plot the magnitude of the armature current I_A as a function of E_A. (You may wish to use MATLAB to create this plot.)

SOLUTION

(a) The rated phase voltage of this generator is 14.4 kV / $\sqrt{3}$ = 8313 V. The base impedance of this generator is

$$Z_{\text{base}} = \frac{3V_{\phi,\text{base}}{}^2}{S_{\text{base}}} = \frac{3(8313\text{ V})^2}{100{,}000{,}000\text{ VA}} = 2.07\ \Omega$$

Therefore,

$$R_A \approx 0\ \Omega \text{ (negligible)}$$

$$X_S = (1.0)(2.07\ \Omega) = 2.07\ \Omega$$

(e) Repeating part *(d)*:

With a **10%** decrease, $E_{A2} = 13{,}372$ V, and

$$\delta_2 = \sin^{-1}\left(\frac{E_{A1}}{E_{A2}}\sin\delta_2\right) = \sin^{-1}\left(\frac{14{,}858\text{ V}}{13{,}372\text{ V}}\sin 26.54°\right) = 29.8°$$

Therefore, the new armature current is

$$\mathbf{I}_A = \frac{\mathbf{E}_{A2} - \mathbf{V}_\phi}{jX_S} = \frac{13{,}372\angle 29.8° - 8313\angle 0°}{j2.07} = 3582\angle -26.3°\text{ A}$$

With a **15%** decrease, $E_{A2} = 12{,}629$ V, and

$$\delta_2 = \sin^{-1}\left(\frac{E_{A1}}{E_{A2}}\sin\delta_2\right) = \sin^{-1}\left(\frac{14{,}858\text{ V}}{12{,}629\text{ V}}\sin 26.54°\right) = 31.7°$$

Therefore, the new armature current is

$$\mathbf{I}_A = \frac{\mathbf{E}_{A2} - \mathbf{V}_\phi}{jX_S} = \frac{12{,}629\angle 31.7° - 8313\angle 0°}{j2.07} = 3414\angle -20.1°\text{ A}$$

With a **20%** decrease, $E_{A2} = 11{,}886$ V, and

$$\delta_2 = \sin^{-1}\left(\frac{E_{A1}}{E_{A2}}\sin\delta_2\right) = \sin^{-1}\left(\frac{14{,}858\text{ V}}{11{,}886\text{ V}}\sin 26.54°\right) = 34.0°$$

Therefore, the new armature current is

$$\mathbf{I}_A = \frac{\mathbf{E}_{A2} - \mathbf{V}_\phi}{jX_S} = \frac{11{,}886\angle 34.0° - 8313\angle 0°}{j2.07} = 3296\angle -13.1°\text{ A}$$

With a **25%** decrease, $E_{A2} = 11{,}144$ V, and

$$\delta_2 = \sin^{-1}\left(\frac{E_{A1}}{E_{A2}}\sin\delta_2\right) = \sin^{-1}\left(\frac{14{,}858\text{ V}}{11{,}144\text{ V}}\sin 26.54°\right) = 36.6°$$

Therefore, the new armature current is

$$\mathbf{I}_A = \frac{\mathbf{E}_{A2} - \mathbf{V}_\phi}{jX_S} = \frac{11{,}144\angle 36.6° - 8313\angle 0°}{j2.07} = 3224\angle -5.4°\text{ A}$$

5-3. A 230-V, 50 Hz, two-pole synchronous motor draws 40 A from the line at unity power factor and full load. Assuming that the motor is lossless, answer the following questions:

(a) What is the output torque of this motor? Express the answer both in newton-meters and in pound-feet.

(b) What must be done to change the power factor to 0.85 leading? Explain your answer, using phasor diagrams.

(c) What will the magnitude of the line current be if the power factor is adjusted to 0.85 leading?

SOLUTION

(a) If this motor is assumed lossless, then the input power is equal to the output power. The input power to this motor is

$$P_{\text{IN}} = \sqrt{3} V_T I_L \cos\theta = \sqrt{3}(230\text{ V})(40\text{ A})(1.0) = 15.93\text{ kW}$$

The rotational speed of the motor is

$$n_m = \frac{120 f_{se}}{P} = \frac{120(50\text{ Hz})}{4} = 1500\text{ r/min}$$

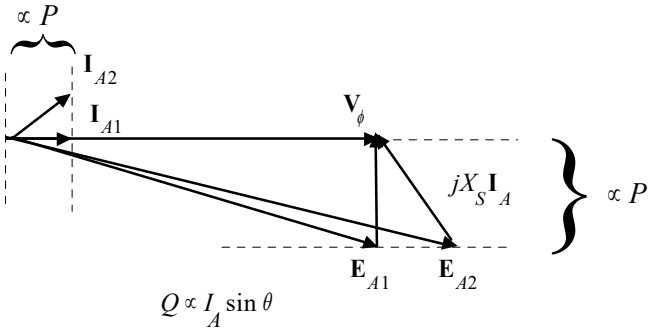
The output torque would be

$$\tau_{\text{LOAD}} = \frac{P_{\text{OUT}}}{\omega_m} = \frac{15.93\text{ kW}}{(1500\text{ r/min})\left(\frac{1\text{ min}}{60\text{ s}}\right)\left(\frac{2\pi\text{ rad}}{1\text{ r}}\right)} = 101.4\text{ N}\cdot\text{m}$$

In English units,

$$\tau_{\text{LOAD}} = \frac{7.04\, P_{\text{OUT}}}{n_m} = \frac{(7.04)(15.93\text{ kW})}{(1500\text{ r/min})} = 74.8\text{ lb}\cdot\text{ft}$$

(b) To change the motor's power factor to 0.8 leading, its field current must be increased. Since the power supplied to the load is independent of the field current level, an increase in field current increases $|\mathbf{E}_A|$ while keeping the distance $E_A \sin\delta$ constant. This increase in E_A changes the angle of the current $\mathbf{I}_A$, eventually causing it to reach a power factor of 0.8 leading.



(c) The magnitude of the line current will be

$$I_L = \frac{P}{\sqrt{3}\ V_T\ \text{PF}} = \frac{15.93\text{ kW}}{\sqrt{3}(230\text{ V})(0.8)} = 50.0\text{ A}$$

5-8. A 4.12 kV, 60 Hz, 3000-hp 0.8-PF-leading, Δ-connected, three-phase synchronous motor has a synchronous reactance of 1.1 per unit and an armature resistance of 0.1 per unit. If this motor is running at rated voltage with a line current of 300 A at 0.85 PF leading, what is the internal generated voltage per phase inside this motor? What is the torque angle δ?

SOLUTION

The output power of the motor is $(3000 \text{ hp})(746 \text{ W/hp}) = 2238 \text{ kW}$. If we take this as rated power, the ratings of this machine are

$$S_{\text{base}} = P / \text{PF} = (2238 \text{ kW}) / (0.8) = 2798 \text{ kVA}$$

$$V_{L,\text{base}} = 4120 \text{ V}$$

$$V_{\phi,\text{base}} = 4120 \text{ V}$$

$$I_{L,\text{base}} = \frac{S_{\text{base}}}{\sqrt{3} V_{L,\text{base}}} = \frac{2798 \text{ kVA}}{\sqrt{3}(4120 \text{ V})} = 392 \text{ A}$$

$$I_{\phi,\text{base}} = \frac{I_{L,\text{base}}}{\sqrt{3}} = \frac{392 \text{ A}}{\sqrt{3}} = 226 \text{ A}$$

Therefore, the line current of 300 A in per-unit is

$$I_{L,\text{pu}} = \frac{I_L}{I_{L,\text{base}}} = \frac{300 \text{ A}}{392 \text{ A}} = 0.765 \text{ pu}$$

and the final per-unit current is

$$I_{\text{pu}} = 0.765 \angle 31.8° \text{ pu}$$

The internal generated voltage in per-unit is

$$\mathbf{E}_A = \mathbf{V}_\phi - R_A \mathbf{I}_A - jX_S \mathbf{I}_A$$

$$\mathbf{E}_A = 1\angle 0° - (0.1)(0.765\angle 31.8°) - j(1.1)(0.765\angle 31.8°)$$

$$\mathbf{E}_A = 1572 \angle -28.7° \text{ pu}$$

In volts, the internal generated voltage is

$$\mathbf{E}_A = (1.572 \angle -28.7° \text{ pu})(4120 \text{ V}) = 6477 \angle -28.7° \text{ V}$$

And the torque angle δ is -28.7 .

5-10. A synchronous machine has a synchronous reactance of 1.0 Ω per phase and an armature resistance of 0.1 Ω per phase. If $\mathbf{E}_A$ = 460∠-10° V and $\mathbf{V}_\phi$ = 480∠0° V, is this machine a motor or a generator? How much power *P* is this machine consuming from or supplying to the electrical system? How much reactive power *Q* is this machine consuming from or supplying to the electrical system?

SOLUTION This machine is a motor, *consuming* power from the power system, because $\mathbf{E}_A$ is lagging $\mathbf{V}_\phi$. It is also *consuming* reactive power, because $E_A \cos\delta < V_\phi$. The current flowing in this machine is

$$\mathbf{I}_A = \frac{\mathbf{V}_\phi - \mathbf{E}_A}{R_A + jX_S} = \frac{480\angle 0° \text{ V} - 460\angle -10° \text{ V}}{0.1 + j1.0\ \Omega} = 83.9\angle -13° \text{ A}$$

Therefore the real power consumed by this motor is

$$P = 3V_\phi I_A \cos\theta = 3(480 \text{ V})(89.3 \text{ A})\cos(13°) = 125.3 \text{ kW}$$

and the reactive power consumed by this motor is

$$Q = 3V_\phi I_A \sin\theta = 3(480 \text{ V})(89.3 \text{ A})\sin(13°) = 28.9 \text{ kVAR}$$

5-11. A 500 kVA, 600 V, 0.8-PF-leading, Y-connected synchronous motor has a synchronous reactance of 1.0 per unit and an armature resistance of 0.1 per unit. At the current time, $\mathbf{E}_A$ = 1.00∠12° pu and $\mathbf{V}_\phi$ = 1∠0° pu.

(a) Is this machine currently acting as a motor or a generator?

(b) How much power *P* is this machine consuming from or supplying to the electrical system?

(c) How much reactive power *Q* is this machine consuming from or supplying to the electrical system?

(d) Is this machine operating within its rated limits?

SOLUTION

(a) This machine is a generator, *supplying* power to the power system, because $\mathbf{E}_A$ is leading $\mathbf{V}_\phi$.

It is also *consuming* reactive power, because $E_A \cos\delta < V_\phi$.

(b) The per-unit current flowing in this machine (assuming that it is a generator) is

$$\mathbf{E}_A = \mathbf{V}_\phi + R_A\mathbf{I}_A + jX_S\mathbf{I}_A$$

$$\mathbf{I}_A = \frac{\mathbf{E}_A - \mathbf{V}_\phi}{R_A + jX_S} = \frac{1\angle 12° \text{ pu} - 1\angle 0° \text{ pu}}{0.1 + j1.0} = 0.208\angle 11.7° \text{ A}$$

The current angle in this generator is 11.7°, so the impedance angle θ is -11.7°. Therefore the real power supplied to the power system by this machine is

$$P = 3V_\phi I_A \cos\theta = 3(1.0)(0.208)\cos(-11.7°) = 0.611 \text{ pu}$$

Converting from per-unit to real power in watts, we get

$$P = S_{\text{base}} P_{\text{pu}} = (500 \text{ kVA})(0.611 \text{ pu}) = 305.5 \text{ kW}$$

(c) The reactive power consumed by this motor is

$$Q = 3V_\phi I_A \sin\theta = 3(1.0)(0.208)\sin(-11.7°) = -0.127 \text{ pu}$$

Converting from per-unit to reactive power in var, we get

$$Q = S_{\text{base}} Q_{\text{pu}} = (500 \text{ kVA})(-0.127 \text{ pu}) = -63.5 \text{ kvar}$$

(d) The total apparent power of this machine is

$$S = \sqrt{P^2 + Q^2} = \sqrt{(305.5 \text{ kW})^2 + (63.5 \text{ kvar})^2} = 312 \text{ kVA}$$

The machine is operating within limits.

5-12. Figure P5-3 shows a small industrial plant supplied by an external 480 V three-phase power supply. The plant includes three main loads as shown in the figure. Answer the following questions about the plant. The synchronous motor is rated at 100 hp, 460 V, and 0.8-PF-leading. The synchronous reactance is 1.1 pu and armature resistance is 0.01 pu. The OCC for this motor is shown in Figure P5-4.

(a) If the switch on the synchronous motor is open, how much real, reactive, and apparent power is being supplied to the plant? What is the current I_L in the transmission line?

The switch is now closed and the synchronous motor is supplying rated power at rated power factor.

***(b)* What is the field current in the motor?**

(c) What is the torque angle of the motor?

(c) What is the power factor of the motor?

(d) How much real, reactive, and apparent power is being supplied to the plant now? What is the current I_L in the transmission line?

Now suppose that the field current is increased to 3.0 A.

(e) What is the real and reactive power supplied to the motor?

(f) What is the torque angle of the motor?

(g) What is the power factor of the motor?

(h) How much real, reactive, and apparent power is being supplied to the plant now? What is the current I_L in the transmission line?

(i) **How does the line current in part *(d)* compare to the line current in part *(h)*? Why?**

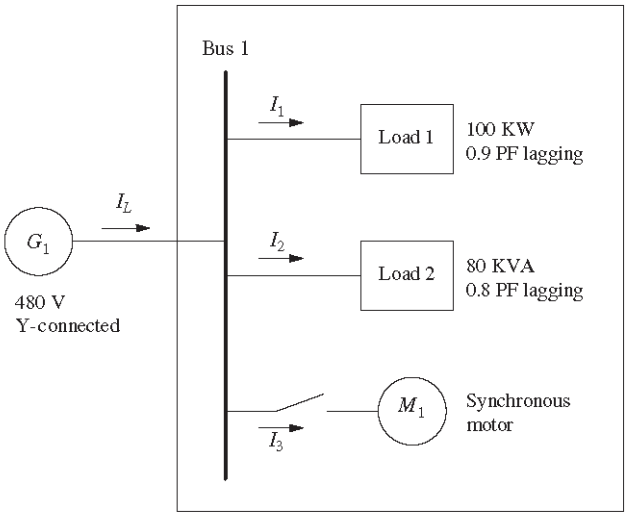


FIGURE P5–3
A small industrial facility.

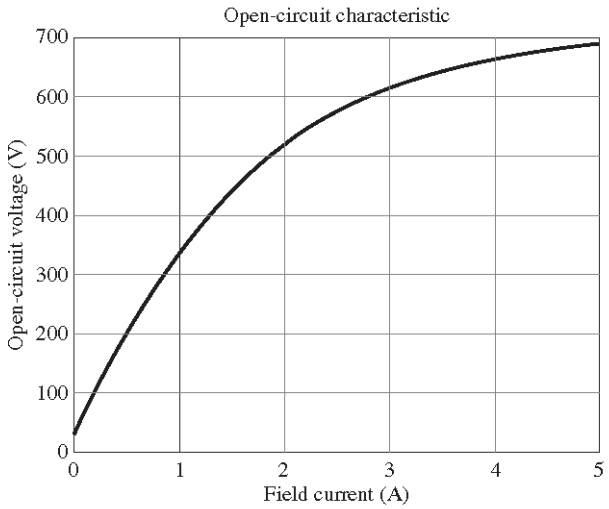


FIGURE P5–4
Open-circuit characteristic of synchronous motor.

SOLUTION

(a) The real and reactive power supplied by Load 1 is

$$P_1 = 100 \text{ kW}$$

$$Q_1 = P_1 \tan\left(\cos^{-1} \text{PF}\right) = \left(100 \text{ kW}\right) \tan\left(\cos^{-1}\left(0.9\right)\right) = 48.4 \text{ kvar}$$

The real and reactive power supplied by Load 2 is

$$P_2 = S_2 \cos\theta = \left(80 \text{ kVA}\right)\left(0.8\right) = 64 \text{ kW}$$

$$I_A = \frac{P_{motor}}{3V_\phi \cos\theta} = \frac{74.6 \text{ kW}}{3\left(277 \text{ V}\right)\left(0.8\right)} = 112 \text{ A}$$

Since the power factor is 0.8 leading, the current $\mathbf{I}_A = 112\angle 36.87° \text{ A}$, and the internal generated voltage is

$$\mathbf{E}_A = \mathbf{V}_\phi - R_A\mathbf{I}_A - jX_S\mathbf{I}_A$$

$$\mathbf{E}_A = \left(277\angle 0° \text{ V}\right) - \left(0.023 \ \Omega\right)\left(112\angle 36.87° \text{ A}\right) - j\left(2.51 \ \Omega\right)\left(112\angle 36.87° \text{ A}\right)$$

$$\mathbf{E}_A = 498\angle -27.0° \text{ pu}$$

The field current must corresponding to 498 V is 1.86 A.

(c) The torque angle of the motor is $\delta = -27°$

(d) The real, reactive, and apparent power supplied to the plant is

$$P_{\text{TOT}} = P_1 + P_2 + P_{motor} = 100 \text{ kW} + 64 \text{ kW} + 74.6 \text{ kW} = 238.6 \text{ kW}$$

$$Q_{\text{TOT}} = Q_1 + Q_2 + Q_{motor} = 48.4 \text{ kvar} + 48 \text{ kvar} - 56 \text{ kvar} = 40.4 \text{ kvar}$$

$$S_{\text{TOT}} = \sqrt{{P_{\text{TOT}}}^2 + {Q_{\text{TOT}}}^2} = \sqrt{\left(238.6 \text{ kW}\right)^2 + \left(40.4 \text{ kVA}\right)^2} = 242 \text{ kVA}$$

The line current is given by

$$I_L = \frac{S_{\text{TOT}}}{\sqrt{3}V_L} = \frac{242 \text{ kVA}}{\sqrt{3}\left(480 \text{ V}\right)} = 291 \text{ A}$$

(e) If the field current is increased to 2 A, the magnitude of the internal generated voltage will rise from 498 V to 517 V (see the OCC curve). The power supplied to the load will remain constant (because the load torque and rotational speed are unchanged), so the distance $E_A \sin\delta \propto P$ will remain constant. Therefore the torque angle becomes

$$\delta_2 = \sin^{-1}\left(\frac{E_{A1}}{E_{A2}}\sin\delta_1\right) = \sin^{-1}\left(\frac{498\text{ V}}{517\text{ V}}\sin(-27°)\right) = -25.9°$$

The new armature current is

$$\mathbf{I}_{A2} = \frac{\mathbf{V}_\phi - \mathbf{E}_{A2}}{jX_S} = \frac{277\angle 0°\text{ V} - 517\angle -25.9°\text{ V}}{j2.51\,\Omega} = 117\angle 39.8°\text{ A}$$

The current angle is 39.8°, so the impedance angle $\theta = -39.8°$. The real power supplied by the motor is

$$P_{motor} = 3V_\phi I_\phi \cos\theta = 3(277\text{ V})(117\text{ A})\cos(-39.8°) = 74.6\text{ kW}$$

$$Q_{motor} = 3V_\phi I_\phi \cos\theta = 3(277\text{ V})(117\text{ A})\sin(-39.8°) = -62.2\text{ kvar}$$

(f) The torque angle of the motor is $\delta = -39.8°$

(g) The power factor of the motor is $\cos(-39.8°) = 0.768$ leading.

(h) The real, reactive, and apparent power supplied to the plant is

$$P_{\text{TOT}} = P_1 + P_2 + P_{motor} = 100\text{ kW} + 64\text{ kW} + 74.6\text{ kW} = 238.6\text{ kW}$$

$$Q_{\text{TOT}} = Q_1 + Q_2 + Q_{motor} = 48.4\text{ kvar} + 48\text{ kvar} - 62.2\text{ kvar} = 26.2\text{ kvar}$$

$$S_{\text{TOT}} = \sqrt{{P_{\text{TOT}}}^2 + {Q_{\text{TOT}}}^2} = \sqrt{(238.6\text{ kW})^2 + (26.2\text{ kVA})^2} = 240\text{ kVA}$$

The line current is given by

$$I_L = \frac{S_{\text{TOT}}}{\sqrt{3}V_L} = \frac{240\text{ kVA}}{\sqrt{3}(480\text{ V})} = 288.7\text{ A}$$

(i) The line current in part *(d)* is greater than the line current in part *(h)*, because the synchronous motor is supplying more of the reactive power being consumed by the other two loads, required less reactive power from the infinite bus.

5-14. A 2300-V, 400-hp, 60-Hz, eight-pole, Y-connected synchronous motor has a rated power factor of 0.85 leading. At full load, the efficiency is 90 percent. The armature resistance is 0.8 Ω, and the synchronous reactance is 11 Ω. Find the following quantities for this machine when it is operating at full load:

(a) Output torque

(b) Input power

(c) n_m

(d) $\mathbf{E}_A$

(e) $|\mathbf{I}_A|$

(f) P_{conv}

(g) $P_{\text{mech}} + P_{\text{core}} + P_{\text{stray}}$

SOLUTION

(a) Since this machine has 8 poles, it rotates at a speed of

$$n_m = \frac{120 f_{se}}{P} = \frac{120(60 \text{ Hz})}{8} = 900 \text{ r/min}$$

At full load, the power supplied by this machine is

$$P_{\text{out}} = (400 \text{ hp})(746 \text{ W/hp}) = 298 \text{ kW}$$

If the output power is 298 kW, the output torque is

$$\tau_{\text{load}} = \frac{P_{\text{out}}}{\omega_{\text{m}}} = \frac{(298{,}000 \text{ W})}{(900 \text{ r/min})\left(\frac{2\pi \text{ rad}}{1 \text{ r}}\right)\left(\frac{1 \text{ min}}{60 \text{ s}}\right)} = 3162 \text{ N} \cdot \text{m}$$

(b) The input power is

$$P_{\text{IN}} = \frac{P_{\text{OUT}}}{\eta} = \frac{298 \text{ kW}}{0.90} = 331 \text{ kW}$$

(c) The mechanical speed is

$$n_m = 900 \text{ r/min}$$

(d) The armature current is

$$I_A = I_L = \frac{P_{\text{IN}}}{\sqrt{3}\, V_T \text{ PF}} = \frac{331 \text{ kW}}{\sqrt{3}(2300 \text{ V})(0.85)} = 97.8 \text{ A}$$

$$\mathbf{I}_A = 97.8\angle 31.8° \text{ A}$$

The phase voltage is $\mathbf{V}_\phi = (2300 \text{ V})/\sqrt{3}\angle 0° = 1328\angle 0° \text{ V}$. Therefore, $\mathbf{E}_A$ is

$$\mathbf{E}_A = \mathbf{V}_\phi - R_A \mathbf{I}_A - jX_S \mathbf{I}_A$$
$$\mathbf{E}_A = (1328\angle 0° \text{ V}) - (0.8 \Omega)(97.8\angle 31.8° \text{ A}) - j(11 \Omega)(97.8\angle 31.8° \text{ A})$$
$$\mathbf{E}_A = 2063\angle -27.6° \text{ V}$$

(e) The magnitude of the armature current is 97.8 A.

(f) The power converted from electrical to mechanical form is given by the equation $P_{\text{conv}} = P_{\text{IN}} - P_{\text{CU}}$

$$P_{\text{CU}} = 3I_A^2 R_A = 3(97.8 \text{ A})^2 (0.8 \Omega) = 23 \text{ kW}$$

$$P_{\text{conv}} = P_{\text{IN}} - P_{\text{CU}} = 331 \text{ kW} - 23 \text{ kW} = 308 \text{ kW}$$

(g) The mechanical, core, and stray losses are given by the equation

$$P_{\text{mech}} + P_{\text{core}} + P_{\text{stray}} = P_{\text{conv}} - P_{\text{OUT}} = 308 \text{ kW} - 298 \text{ kW} = 10 \text{ kW}$$

5-16. A 480-V, 500-kVA, 0.8-PF-lagging, Y-connected synchronous generator has a synchronous reactance of 0.4 Ω and a negligible armature resistance. This generator is supplying power to a 480-V, 80-kW, 0.8-PF-leading, Y-connected synchronous motor with a synchronous reactance of 2.0 Ω and a negligible armature resistance. The synchronous generator is adjusted to have a terminal voltage of 480 V when the motor is drawing the rated power at unity power factor.

(a) Calculate the magnitudes and angles of $\mathbf{E}_A$ for both machines.

(b) If the flux of the motor is increased by 10 percent, what happens to the terminal voltage of the power system? What is its new value?

(c) What is the power factor of the motor after the increase in motor flux?

SOLUTION

(a) The motor is Y-connected, and is operating at rated voltage (480 V), rated power (80 kW), and unity power factor. The current flowing in the motor is

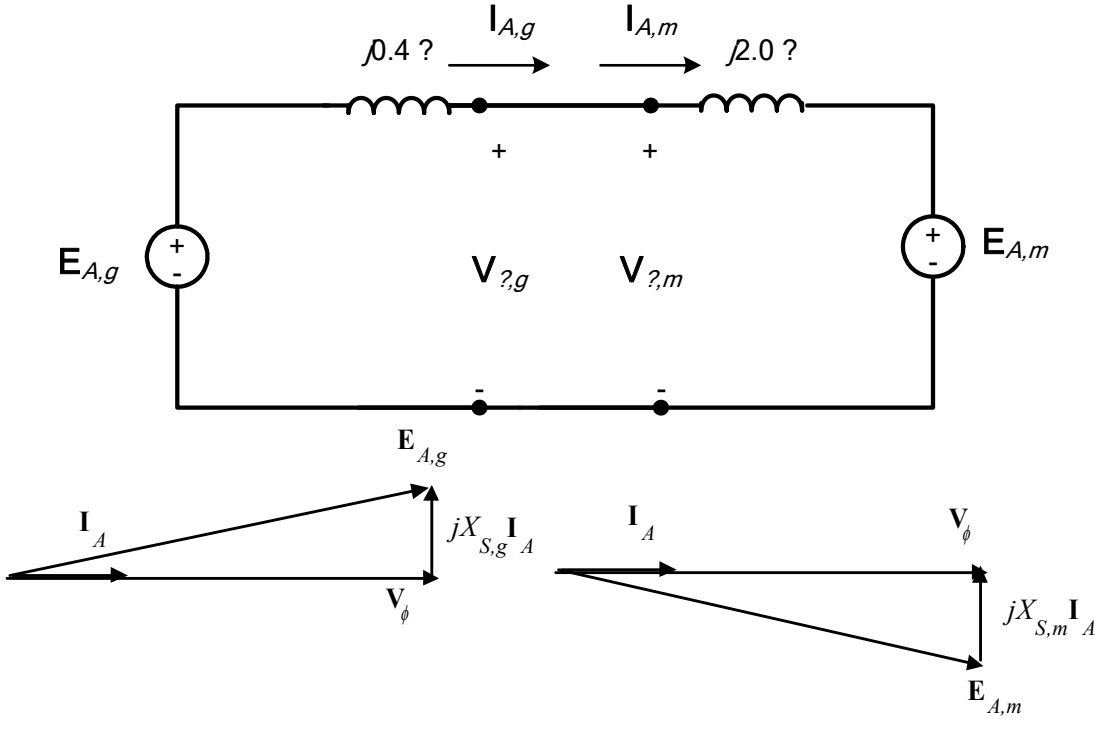
$$I_{A,\text{m}} = I_{L,\text{m}} = \frac{P}{\sqrt{3}\ V_T\ \text{PF}} = \frac{80\ \text{kW}}{\sqrt{3}(480\ \text{V})(1.0)} = 96.2\ \text{A}$$

so $\mathbf{I}_{A,\text{m}} = 96.2\angle 0°\ \text{A}$. This machine is Y-connected, so the phase voltage is $V_\phi = 480 / \sqrt{3} = 277$ V. The internal generated voltage of the motor is

$$\mathbf{E}_{A,\text{m}} = \mathbf{V}_{\phi,\text{m}} - jX_S\mathbf{I}_{A,\text{m}}$$
$$\mathbf{E}_{A,\text{m}} = 277\angle 0°\ \text{V} - j(2.0\ \Omega)(96.2\angle 0°\ \text{A})$$
$$\mathbf{E}_{A,\text{m}} = 337\angle -34.8°\ \text{V}$$

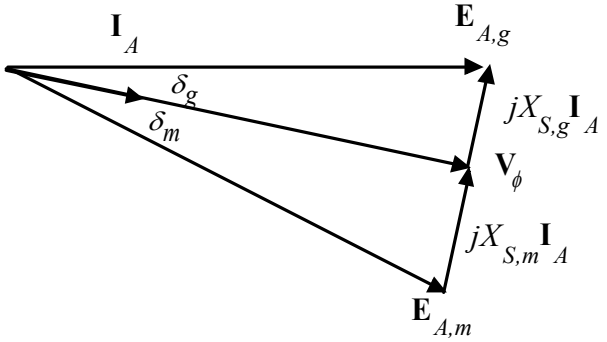
The current supplied to the motor comes from the generator, so the internal generated voltage of the generator is

$$\mathbf{E}_{A,\text{g}} = \mathbf{V}_{\phi,\text{g}} + jX_S\mathbf{I}_{A,\text{g}}$$
$$\mathbf{E}_{A,\text{g}} = 277\angle 0°\ \text{V} + j(0.4\ \Omega)(96.2\angle 0°\ \text{A})$$
$$\mathbf{E}_{A,\text{g}} = 280\angle 7.9°\ \text{V}$$



(b) The power supplied by the generator to the motor will be constant as the field current of the motor is varied, because the power is a function of the load on the motor, and that load has not changed. The 10% increase in flux will raise the internal generated voltage of the motor to (1.1)(337 V) = 371 V.

To make finding the new conditions easier, *we will make the angle of the phasor* $\mathbf{E}_{A,g}$ *the reference phase angle during the following calculations*. The resulting phasor diagram is shown below. Note that $jX_{S,g}\mathbf{I}_A$ and $jX_{S,m}\mathbf{I}_A$ are at the same angle, so they just add together.



Then by Kirchhoff's Voltage Law,

$$\mathbf{E}_{A,g} = \mathbf{E}_{A,m} + j(X_{S,g} + X_{S,m})\mathbf{I}_A$$

or $$\mathbf{I}_A = \frac{\mathbf{E}_{A,g} - \mathbf{E}_{A,m}}{j(X_{S,g} + X_{S,m})}$$

Note that this combined phasor diagram looks just like the diagram of a synchronous motor, so we can apply the power equation for synchronous motors to this system.

$$P = \frac{3E_{A,g}E_{A,m}}{X_{S,g} + X_{S,m}} \sin\gamma$$

where $\gamma = \delta_g + \delta_m$. From this equation,

$$\gamma = \sin^{-1}\frac{(X_{S,g} + X_{S,m})P}{3E_{A,g}E_{A,m}} = \sin^{-1}\frac{(2.4\ \Omega)(80\ \text{kW})}{3(280\text{V})(371\ \text{V})} = 38.0°$$

Therefore,

$$\mathbf{I}_A = \frac{\mathbf{E}_{A,g} - \mathbf{E}_{A,m}}{j(X_{S,g} + X_{S,m})} = \frac{280\angle 0°\ \text{V} - 371\angle -38.0°\ \text{V}}{j2.4\ \Omega} = 95.3\angle 3.1°\ \text{A}$$

The phase voltage of the system would be

$$\mathbf{V}_\phi = \mathbf{E}_{A,g} - jX_{S,g}\mathbf{I}_A = 280\angle 0°\ \text{V} - j(0.4\ \Omega)(95.3\angle 3.1°\ \text{A}) = 285\angle -7.7°\ \text{V}$$

The phase voltage of the system can also be calculated as

$$\mathbf{V}_\phi = \mathbf{E}_{A,g} - jX_{S,m}\mathbf{I}_A = 371\angle -38°\ \text{V} + j(2\ \Omega)(95.3\angle 3.1°\ \text{A}) = 284.6\angle -7.7°\ \text{V}$$

These two calculations agree, as we would expect.

If we now make $\mathbf{V}_\phi$ the reference (as we usually do), all of the phases will shift by 7.7°, and these voltages and currents become:

$$\mathbf{E}_{A,g} = 280\angle 7.7°\ \text{V}$$
$$\mathbf{V}_\phi = 285\angle 0°\ \text{V}$$
$$\mathbf{E}_{A,m} = 371\angle -30.3°\ \text{V}$$
$$\mathbf{I}_A = 95.3\angle 10.8°\ \text{A}$$

The new terminal voltage is $V_T = \sqrt{3}(285\ \text{V}) = 494\ \text{V}$, so the system voltage has *increased*.

(c) The impedance angle is $\theta = -10.8°$ (the opposite of the current angle). The power factor of the motor is now $\text{PF} = \cos(-10.8°) = 0.982$ leading.

Note: The reactive power in the motor is now

$$Q_{\text{motor}} = 3V_\phi I_A \sin\theta = 3(280\text{ V})(95.3\text{ A})\sin(-10.8°) = -15\text{ kVAR}$$

The motor is now supplying 15 kVAR to the system. Note that **an increase in machine flux has increased the reactive power supplied by the motor and also raised the terminal voltage of the system**. This is consistent with what we learned about reactive power sharing in Chapter 4.

5-17. A 440-V, 60 Hz, three-phase Y-connected synchronous motor has a synchronous reactance of 1.5 Ω per phase. The field current has been adjusted so that the torque angle δ is 25° when the power supplied by the generator is **80** kW.

(a) What is the magnitude of the internal generated voltage $\mathbf{E}_A$ in this machine?

(b) What are the magnitude and angle of the armature current in the machine? What is the motor's power factor?

(c) If the field current remains constant, what is the absolute maximum power this motor could supply?

SOLUTION

(a) The power supplied to the motor is 80 kW. This power is given by the equation

$$P = \frac{3V_\phi E_A}{X_S}\sin\delta$$

The phase voltage of the generator is $440/\sqrt{3} = 254$ V, so the magnitude of $\mathbf{E}_A$ is

$$E_A = \frac{X_S\ P}{3V_\phi \sin\delta} = \frac{(1.5\ \Omega)(80\text{ kW})}{3(254\text{ V})\sin 25°} = 373\text{ V}$$

(b) The armature current in this machine is given by

$$\mathbf{I}_A = \frac{\mathbf{V}_\phi - \mathbf{E}_A}{jX_S} = \frac{254\angle 0°\text{ V} - 373\angle -25°}{j1.5} = 119\angle 28°\text{ A}$$

The power factor of the motor is PF = cos 28º = 0.883 leading.

(c) The maximum power that the motor could supply at this field current is

$$P_{\max} = \frac{3V_\phi E_A}{X_S} = \frac{3(254\text{ V})(373\text{ V})}{1.5\ \Omega} = 189.4\text{ kW}$$

5-19. A 100-hp 440-V 0.8-PF-leading Δ-connected synchronous motor has an armature resistance of 0.3 Ω and a synchronous reactance of 4.0 Ω. Its efficiency at full load is 96 percent.

(a) What is the input power to the motor at rated conditions?

(b) What is the line current of the motor at rated conditions? What is the phase current of the motor at rated conditions?

(c) What is the reactive power consumed by or supplied by the motor at rated conditions?

(d) What is the internal generated voltage $\mathbf{E}_A$ of this motor at rated conditions?

(e) What are the stator copper losses in the motor at rated conditions?

(f) What is P_{conv} at rated conditions?

(g) If E_A is decreased by 10 percent, how much reactive power will be consumed by or supplied by the motor?

SOLUTION

(a) The input power to the motor at rated conditions is

$$P_{\text{IN}} = \frac{P_{\text{OUT}}}{\eta} = \frac{(100 \text{ hp})(746 \text{ W/hp})}{0.96} = 77.7 \text{ kW}$$

(b) The line current to the motor at rated conditions is

$$I_L = \frac{P}{\sqrt{3}\ V_T\ \text{PF}} = \frac{77.7 \text{ kW}}{\sqrt{3}(440 \text{ V})(0.85)} = 120 \text{ A}$$

The phase current to the motor at rated conditions is

$$I_\phi = \frac{I_L}{\sqrt{3}} = \frac{120 \text{ A}}{\sqrt{3}} = 69.3 \text{ A}$$

(c) The reactive power supplied by this motor to the power system at rated conditions is

$$Q_{\text{rated}} = 3V_\phi I_A \sin\theta = 3(440 \text{ V})(69.3 \text{ A})\sin 36.87° = 54.9 \text{ kVAR}$$

(d) The internal generated voltage at rated conditions is

$$\mathbf{E}_A = \mathbf{V}_\phi - R_A\mathbf{I}_A - jX_S\mathbf{I}_A$$
$$\mathbf{E}_A = 440\angle 0° \text{ V} - (0.3 \Omega)(69.3\angle 36.87° \text{ A}) - j(4.0 \Omega)(69.3\angle 36.87° \text{ A})$$
$$\mathbf{E}_A = 634\angle -21.7° \text{ V}$$

(e) The stator copper losses at rated conditions are

$$P_{\text{CU}} = 3I_A^2 R_A = 3(69.3 \text{ A})^2 (0.3 \Omega) = 4.3 \text{ kW}$$

(f) P_{conv} at rated conditions is

$$P_{\text{conv}} = P_{\text{IN}} - P_{\text{CU}} = 77.7 \text{ kW} - 4.3 \text{ kW} = 73.4 \text{ kW}$$

(g) If E_A is decreased by 10%, the new value if E_A = (0.9)(634 V) = 571 V. To simplify this part of the problem, we will ignore R_A. Then the quantity $E_A \sin\delta$ will be constant as E_A changes. Therefore,

$$\delta_2 = \sin^{-1}\left(\frac{E_{A1}}{E_{A2}}\sin\delta_1\right) = \sin^{-1}\left(\frac{634 \text{ V}}{571 \text{ V}}\sin(-21.7°)\right) = -24.2°$$

Therefore,

$$\mathbf{I}_A = \frac{\mathbf{V}_\phi - \mathbf{E}_A}{jX_S} = \frac{440\angle 0° \text{ V} - 571\angle -24.2°}{j4.0} = 61.9\angle 19° \text{ A}$$

and the reactive power supplied by the motor to the power system will be

$$Q = 3V_\phi I_A \sin\theta = 3(440 \text{ V})(61.9 \text{ A})\sin(19°) = 26.6 \text{ kVAR}$$

5-20. Answer the following questions about the machine of Problem 5-19.

(a) If $\mathbf{E}_A$ = 430∠15° V and $\mathbf{V}_\phi$ = 440∠0° V, is this machine consuming real power from or supplying real power to the power system? Is it consuming reactive power from or supplying reactive power to the power system?

(b) Calculate the real power *P* and reactive power *Q* supplied or consumed by the machine under the conditions in part *(a)*. Is the machine operating within its ratings under these circumstances?

(c) If $\mathbf{E}_A$ = 470∠-20° V and $\mathbf{V}_\phi$ = 440∠0° V, is this machine consuming real power from or supplying real power to the power system? Is it consuming reactive power from or supplying reactive power to the power system?

(d) Calculate the real power *P* and reactive power *Q* supplied or consumed by the machine under the conditions in part *(c)*. Is the machine operating within its ratings under these circumstances?

SOLUTION

(a) This machine is a generator supplying real power to the power system, because $\mathbf{E}_A$ is ahead of $\mathbf{V}_\phi$. It is consuming reactive power because $E_A \cos\delta < V_\phi$.

(b) This machine is acting as a generator, and the current flow in these conditions is

$$\mathbf{I}_A = \frac{\mathbf{E}_A - \mathbf{V}_\phi}{R_A + jX_S} = \frac{430\angle 15° - 440\angle 0° \text{ V}}{0.3 + j4.0} = 28.4\angle 16.8° \text{ A}$$

The real power supplied by this machine is

$$P = 3V_\phi I_A \cos\theta = 3(440 \text{ V})(28.4 \text{ A})\cos(-16.8°) = 35.9 \text{ kW}$$

The reactive power supplied by this machine is

$$Q = 3V_\phi I_A \sin\theta = 3(440 \text{ V})(28.4 \text{ A})\sin(-16.8°) = -10.8 \text{ kVAR}$$

In other words, this machine is consuming reactive power.

(c) This machine is a motor consuming real power from the power system, because $\mathbf{E}_A$ is behind $\mathbf{V}_\phi$. It is supplying reactive power because $E_A \cos\delta > V_\phi$.

(d) This machine is acting as a motor, and the current flow in these conditions is

$$\mathbf{I}_A = \frac{\mathbf{V}_\phi - \mathbf{E}_A}{R_A + jX_S} = \frac{440\angle 0° \text{ V} - 470\angle -20°}{0.3 + j4.0} = 40.1\angle 4.9° \text{ A}$$

The real power consumed by this machine is

$$P = 3V_\phi I_A \cos\theta = 3(440 \text{ V})(40.1 \text{ A})\cos(4.9°) = 52.7 \text{ kW}$$

The reactive power supplied by this machine is

$$Q = 3V_\phi I_A \sin\theta = 3(440 \text{ V})(40.1 \text{ A})\sin(4.9°) = 4.5 \text{ kVAR}$$

In other words, this machine is supplying reactive power. It is operating within its rating limits.

6-5. A 208-V four-pole 60-Hz Y-connected wound-rotor induction motor is rated at 30 hp. Its equivalent circuit components are

$R_1 = 0.100\ \Omega$ $R_2 = 0.070\ \Omega$ $X_M = 10.0\ \Omega$

$X_1 = 0.210\ \Omega$ $X_2 = 0.210\ \Omega$

$P_{\text{mech}} = 500$ W $P_{\text{misc}} \approx 0$ $P_{\text{core}} = 400$ W

For a slip of 0.05, find

(a) The line current I_L

(b) The stator copper losses P_{SCL}

(c) The air-gap power P_{AG}

(d) The power converted from electrical to mechanical form P_{conv}

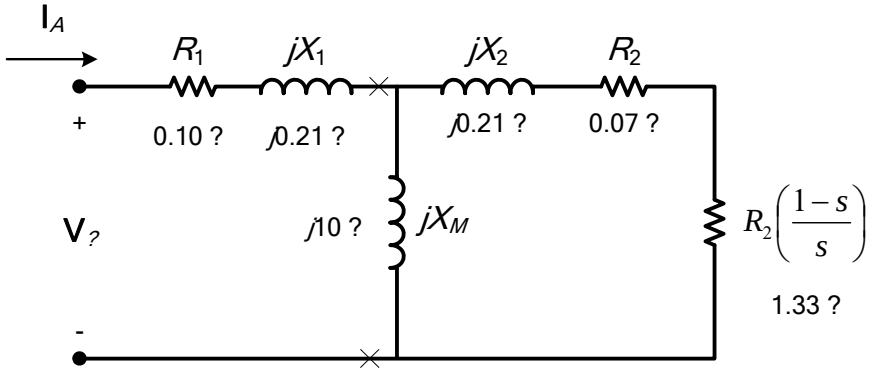
(e) The induced torque τ_{ind}

(f) The load torque τ_{load}

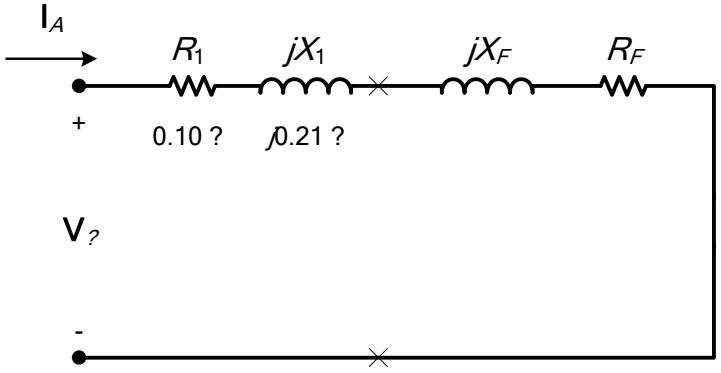
(g) The overall machine efficiency

(h) The motor speed in revolutions per minute and radians per second

SOLUTION The equivalent circuit of this induction motor is shown below:



(a) The easiest way to find the line current (or armature current) is to get the equivalent impedance Z_F of the rotor circuit in parallel with jX_M, and then calculate the current as the phase voltage divided by the sum of the series impedances, as shown below.



The equivalent impedance of the rotor circuit in parallel with jX_M is:

$$Z_F = \frac{1}{\frac{1}{jX_M} + \frac{1}{Z_2}} = \frac{1}{\frac{1}{j10\ \Omega} + \frac{1}{1.40 + j0.21}} = 1.318 + j0.386 = 1.374\angle 16.3°\ \Omega$$

The phase voltage is $208/\sqrt{3} = 120$ V, so line current I_L is

$$I_L = I_A = \frac{V_\phi}{R_1 + jX_1 + R_F + jX_F} = \frac{120\angle 0°\text{ V}}{0.10\ \Omega + j0.21\ \Omega + 1.318\ \Omega + j0.386\ \Omega}$$

$$I_L = I_A = 78.0\angle -22.8°\text{ A}$$

(b) The stator copper losses are

$$P_{\text{SCL}} = 3I_A{}^2 R_1 = 3\left(78.0\text{ A}\right)^2\left(0.10\ \Omega\right) = 1825\text{ W}$$

(c) The air gap power is $P_{\text{AG}} = 3I_2{}^2\dfrac{R_2}{s} = 3I_A{}^2 R_F$

(Note that $3I_A{}^2 R_F$ is equal to $3I_2{}^2\dfrac{R_2}{s}$, since the only resistance in the original rotor circuit was R_2/s, and the resistance in the Thevenin equivalent circuit is R_F. The power consumed by the Thevenin equivalent circuit must be the same as the power consumed by the original circuit.)

$$P_{\text{AG}} = 3I_2{}^2\frac{R_2}{s} = 3I_A{}^2 R_F = 3\left(78.0\text{ A}\right)^2\left(1.318\ \Omega\right) = 24.0\text{ kW}$$

(d) The power converted from electrical to mechanical form is

$$P_{\text{conv}} = \left(1-s\right)P_{\text{AG}} = \left(1-0.05\right)\left(24.0\text{ kW}\right) = 22.8\text{ kW}$$

(e) The induced torque in the motor is

$$\tau_{\text{ind}} = \frac{P_{\text{AG}}}{\omega_{\text{sync}}} = \frac{24.0\text{ kW}}{\left(1800\text{ r/min}\right)\left(\dfrac{2\pi\text{ rad}}{1\text{ r}}\right)\left(\dfrac{1\text{ min}}{60\text{ s}}\right)} = 127.4\text{ N}\cdot\text{m}$$

(f) The output power of this motor is

$$P_{\text{OUT}} = P_{\text{conv}} - P_{\text{mech}} - P_{\text{core}} - P_{\text{misc}} = 22.8\text{ kW} - 500\text{ W} - 400\text{ W} - 0\text{ W} = 21.9\text{ kW}$$

The output speed is

$$n_m = \left(1-s\right)\ n_{\text{sync}} = \left(1-0.05\right)\left(1800\text{ r/min}\right) = 1710\text{ r/min}$$

Therefore the load torque is

$$\tau_{\text{load}} = \frac{P_{\text{OUT}}}{\omega_m} = \frac{21.9\text{ kW}}{\left(1710\text{ r/min}\right)\left(\dfrac{2\pi\text{ rad}}{1\text{ r}}\right)\left(\dfrac{1\text{ min}}{60\text{ s}}\right)} = 122.3\text{ N}\cdot\text{m}$$

(g) The overall efficiency is

$$\eta = \frac{P_{\text{OUT}}}{P_{\text{IN}}}\times 100\% = \frac{P_{\text{OUT}}}{3V_\phi I_A\cos\theta}\times 100\%$$

$$\eta = \frac{21.9\text{ kW}}{3\left(120\text{ V}\right)\left(78.0\text{ A}\right)\cos 22.8^\circ}\times 100\% = 84.6\%$$

(h) The motor speed in revolutions per minute is 1710 r/min. The motor speed in radians per second is

$$\omega_m = \left(1710\text{ r/min}\right)\left(\frac{2\pi\text{ rad}}{1\text{ r}}\right)\left(\frac{1\text{ min}}{60\text{ s}}\right) = 179\text{ rad/s}$$

6-12. The power crossing the air gap of a 60 Hz, four-pole induction motor is 25 kW, and the power converted from electrical to mechanical form in the motor is 23.2 kW.

(a) What is the slip of the motor at this time?

(b) What is the induced torque in this motor?

(c) Assuming that the mechanical losses are 300 W at this slip, what is the load torque of this motor?

SOLUTION

(a) The synchronous speed of this motor is

$$n_{\text{sync}} = \frac{120\, f_{se}}{P} = \frac{120(60\text{ Hz})}{4} = 1800\text{ r/min}$$

The power converted from electrical to mechanical form is

$$P_{\text{conv}} = (1-s)\,P_{\text{AG}}$$

so

$$s = 1 - \frac{P_{\text{conv}}}{P_{\text{AG}}} = 1 - \frac{23.4\text{ kW}}{25\text{ kW}} = 0.064$$

or 6.4%.

(b) The speed of the motor is

$$n_m = (1-s)\,n_{\text{sync}} = (1-0.064)(1800\text{ r/min}) = 1685\text{ r/min}$$

The induced torque of the motor is

$$\tau_{\text{ind}} = \frac{P_{\text{conv}}}{\omega_{\text{m}}} = \frac{23.4\text{ kW}}{(1685\text{ r/min})\left(\frac{2\pi\text{ rad}}{1\text{ r}}\right)\left(\frac{1\text{ min}}{60\text{ s}}\right)} = 132.6\text{ N}\cdot\text{m}$$

Alternately, the induced torque can be found as

$$\tau_{\text{ind}} = \frac{P_{\text{AG}}}{\omega_{\text{sync}}} = \frac{25.0\text{ kW}}{(1800\text{ r/min})\left(\frac{2\pi\text{ rad}}{1\text{ r}}\right)\left(\frac{1\text{ min}}{60\text{ s}}\right)} = 132.6\text{ N}\cdot\text{m}$$

(c) The output power of this motor is

$$P_{\text{out}} = P_{\text{conv}} - P_{\text{mech}} = 23{,}400\text{ W} - 300\text{ W} = 23{,}100\text{ W}$$

$$\tau_{\text{load}} = \frac{P_{\text{out}}}{\omega_{\text{m}}} = \frac{23.1\text{ kW}}{(1685\text{ r/min})\left(\frac{2\pi\text{ rad}}{1\text{ r}}\right)\left(\frac{1\text{ min}}{60\text{ s}}\right)} = 130.9\text{ N}\cdot\text{m}$$

6-15. A 460-V 60-Hz four-pole Y-connected induction motor is rated at 25 hp. The equivalent circuit parameters are

$R_1 = 0.15\ \Omega$ $R_2 = 0.154\ \Omega$ $X_M = 20\ \Omega$

$X_1 = 0.852\ \Omega$ $X_2 = 1.066\ \Omega$

$P_{F\&W} = 400$ W $P_{misc} = 150$ W $P_{core} = 400$ W

For a slip of 0.02, find

(a) The line current I_L

(b) The stator power factor

(c) The rotor power factor

(d) The rotor frequency

(e) The stator copper losses P_{SCL}

(f) The air-gap power P_{AG}

(g) The power converted from electrical to mechanical form P_{conv}

(h) The induced torque τ_{ind}

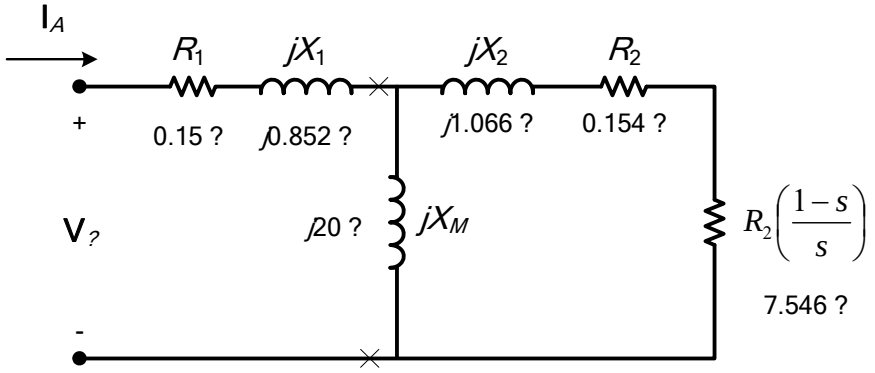
(i) The load torque τ_{load}

(j) The overall machine efficiency η

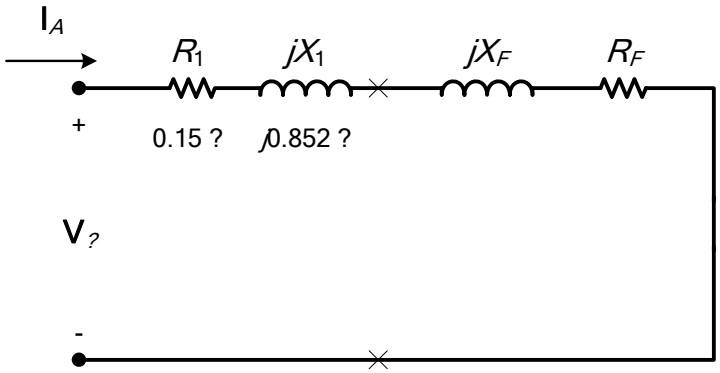
(k) The motor speed in revolutions per minute and radians per second

(l) What is the starting code letter for this motor?

SOLUTION The equivalent circuit of this induction motor is shown below:



(a) The easiest way to find the line current (or armature current) is to get the equivalent impedance Z_F of the rotor circuit in parallel with jX_M, and then calculate the current as the phase voltage divided by the sum of the series impedances, as shown below.



The equivalent impedance of the rotor circuit in parallel with jX_M is:

$$Z_F = \frac{1}{\frac{1}{jX_M} + \frac{1}{Z_2}} = \frac{1}{\frac{1}{j20\ \Omega} + \frac{1}{7.70 + j1.066}} = 6.123 + j3.25 = 6.932\angle 28.0°\ \Omega$$

The phase voltage is $460/\sqrt{3}$ = 266 V, so line current I_L is

$$I_L = I_A = \frac{V_\phi}{R_1 + jX_1 + R_F + jX_F} = \frac{266\angle 0°\text{ V}}{0.15\ \Omega + j0.852\ \Omega + 6.123\ \Omega + j3.25\ \Omega}$$

$$I_L = I_A = 35.5\angle -33.2°\text{ A}$$

(b) The stator power factor is

$$\text{PF} = \cos(33.2°) = 0.837 \text{ lagging}$$

(c) To find the rotor power factor, we must find the impedance angle of the rotor

$$\theta_R = \tan^{-1}\frac{X_2}{R_2/s} = \tan^{-1}\frac{1.066}{7.70} = 7.88°$$

(d) The rotor frequency is

$$f_r = sf_s = (0.02)(60\text{ Hz}) = 1.2\text{ Hz}$$

Therefore the rotor power factor is

$$\text{PF}_R = \cos 7.88° = 0.991 \text{ lagging}$$

(e) The stator copper losses are

$$P_{\text{SCL}} = 3I_A{}^2 R_1 = 3\left(35.5\text{ A}\right)^2\left(0.15\ \Omega\right) = 567\text{ W}$$

(f) The air gap power is $P_{\text{AG}} = 3I_2{}^2\dfrac{R_2}{s} = 3I_A{}^2 R_F$

(Note that $3I_A{}^2 R_F$ is equal to $3I_2{}^2\dfrac{R_2}{s}$, since the only resistance in the original rotor circuit was R_2/s, and the resistance in the Thevenin equivalent circuit is R_F. The power consumed by the Thevenin equivalent circuit must be the same as the power consumed by the original circuit.)

$$P_{\text{AG}} = 3I_2{}^2\frac{R_2}{s} = 3I_A{}^2 R_F = 3\left(35.5\text{ A}\right)^2\left(6.123\ \Omega\right) = 23.15\text{ kW}$$

(g) The power converted from electrical to mechanical form is

$$P_{\text{conv}} = \left(1-s\right)P_{\text{AG}} = \left(1-0.02\right)\ \left(23.15\text{ kW}\right) = 22.69\text{ kW}$$

(h) The synchronous speed of this motor is

$$n_{\text{sync}} = \frac{120\,f_{se}}{P} = \frac{120\left(60\text{ Hz}\right)}{4} = 1800\text{ r/min}$$

$$\omega_{\text{sync}} = \left(1800\text{ r/min}\right)\left(\frac{2\pi\text{ rad}}{1\text{ r}}\right)\left(\frac{1\text{ min}}{60\text{ s}}\right) = 188.5\text{ rad/s}$$

Therefore the induced torque in the motor is

$$\tau_{\text{ind}} = \frac{P_{\text{AG}}}{\omega_{\text{sync}}} = \frac{23.15\text{ kW}}{\left(1800\text{ r/min}\right)\left(\dfrac{2\pi\text{ rad}}{1\text{ r}}\right)\left(\dfrac{1\text{ min}}{60\text{ s}}\right)} = 122.8\text{ N}\cdot\text{m}$$

(i) The output power of this motor is

$$P_{\text{OUT}} = P_{\text{conv}} - P_{\text{mech}} - P_{\text{core}} - P_{\text{misc}} = 22.69\text{ kW} - 400\text{ W} - 400\text{ W} - 150\text{ W} = 21.74\text{ kW}$$

The output speed is

$$n_m = \left(1-s\right)\ n_{\text{sync}} = \left(1-0.02\right)\ \left(1800\text{ r/min}\right) = 1764\text{ r/min}$$

Therefore the load torque is

$$\tau_{\text{load}} = \frac{P_{\text{OUT}}}{\omega_m} = \frac{21.74\text{ kW}}{\left(1764\text{ r/min}\right)\left(\dfrac{2\pi\text{ rad}}{1\text{ r}}\right)\left(\dfrac{1\text{ min}}{60\text{ s}}\right)} = 117.7\text{ N}\cdot\text{m}$$

(j) The overall efficiency is

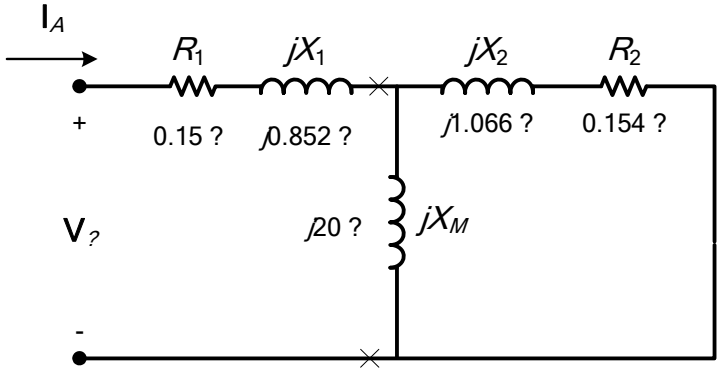
$$\eta = \frac{P_{\text{OUT}}}{P_{\text{IN}}}\times 100\% = \frac{P_{\text{OUT}}}{3V_\phi I_A\cos\theta}\times 100\%$$

$$\eta = \frac{21.74\text{ kW}}{3\left(266\text{ V}\right)\left(35.5\text{ A}\right)\cos\left(33.2°\right)}\times 100\% = 91.7\%$$

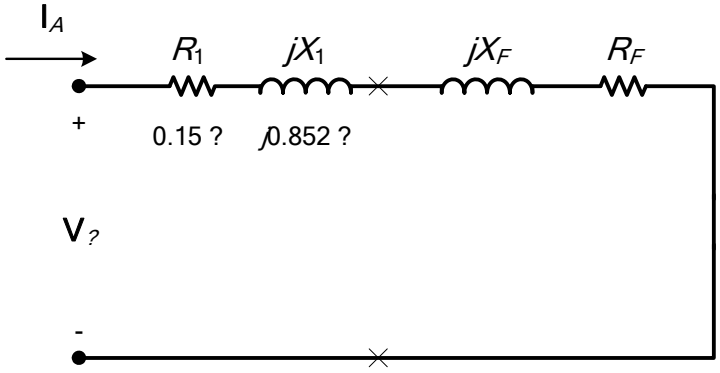
(k) The motor speed in revolutions per minute is 1764 r/min. The motor speed in radians per second is

$$\omega_m = \left(1764\text{ r/min}\right)\left(\frac{2\pi\text{ rad}}{1\text{ r}}\right)\left(\frac{1\text{ min}}{60\text{ s}}\right) = 184.7\text{ rad/s}$$

(l) The equivalent circuit of this induction motor *at starting conditions* is shown below:



The easiest way to find the line current (or armature current) is to get the equivalent impedance Z_F of the rotor circuit in parallel with jX_M, and then calculate the current as the phase voltage divided by the sum of the series impedances, as shown below.



The equivalent impedance of the rotor circuit in parallel with jX_M is:

$$Z_F = \frac{1}{\frac{1}{jX_M} + \frac{1}{Z_2}} = \frac{1}{\frac{1}{j20\ \Omega} + \frac{1}{0.154 + j1.066}} = 0.139 + j1.013 = 1.023\angle 82.2°\ \Omega$$

The phase voltage is $460/\sqrt{3} = 266$ V, so line current I_L is

$$I_L = I_A = \frac{V_\phi}{R_1 + jX_1 + R_F + jX_F} = \frac{266\angle 0°\text{ V}}{0.15\ \Omega + j0.852\ \Omega + 0.139\ \Omega + j1.023\ \Omega}$$

$$I_L = I_A = 140.2\angle -81.2°\text{ A}$$

The starting kVA of the motor is

$$S_{\text{start}} = 3V_\phi I_A = 3(266\text{ V})(140\text{ A}) = 111.7\text{ kVA}$$

The locked rotor kVA/hp is

$$\text{kVA/hp} = \frac{111.7\text{ kVA}}{25\text{ hp}} = 4.47$$

Therefore this motor is Starting Code Letter D.

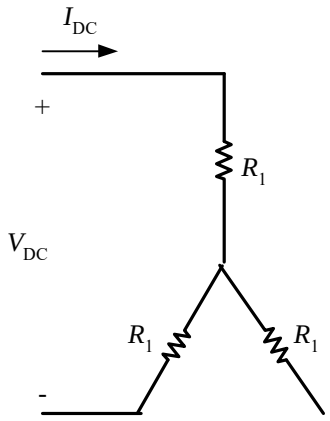
6-20. A 208-V six-pole Y-connected 25-hp design class B induction motor is tested in the laboratory, with the following results:

No load:	208 V, 24.0 A, 1400 W, 60 Hz
Locked rotor:	24.6 V, 64.5 A, 2200 W, 15 Hz
Dc test:	13.5 V, 64 A

Find the equivalent circuit of this motor, and plot its torque-speed characteristic curve.

SOLUTION From the DC test,

$$2R_1 = \frac{13.5\text{ V}}{64\text{ A}} \qquad \Rightarrow \qquad R_1 = 0.105\ \Omega$$



In the no-load test, the line voltage is 208 V, so the phase voltage is 120 V. Therefore,

$$X_1 + X_M = \frac{V_\phi}{I_{A,\text{nl}}} = \frac{120\text{ V}}{24.0\text{ A}} = 5.00\ \Omega \quad @ \quad 60\text{ Hz}$$

In the locked-rotor test, the line voltage is 24.6 V, so the phase voltage is 14.2 V. From the locked-rotor test *at 15 Hz*,

$$|Z'_{\text{LR}}| = |R_{\text{LR}} + jX'_{\text{LR}}| = \frac{V_\phi}{I_{A,\text{LR}}} = \frac{14.2\text{ V}}{64.5\text{ A}} = 0.220\ \Omega$$

$$\theta'_{\text{LR}} = \cos^{-1}\frac{P_{\text{LR}}}{S_{\text{LR}}} = \cos^{-1}\left[\frac{2200\text{ W}}{\sqrt{3}(24.6\text{ V})(64.5\text{ A})}\right] = 36.82°$$

Therefore,

$$R_{\text{LR}} = |Z'_{\text{LR}}|\ \cos\theta_{\text{LR}} = (0.220\ \Omega)\cos(36.82°) = 0.176\ \Omega$$

$$\Rightarrow \qquad R_1 + R_2 = 0.176\ \Omega$$

$$\Rightarrow \qquad R_2 = 0.071\ \Omega$$

$$X'_{\text{LR}} = |Z'_{\text{LR}}|\ \sin\theta_{\text{LR}} = (0.2202\ \Omega)\sin(36.82°) = 0.132\ \Omega$$

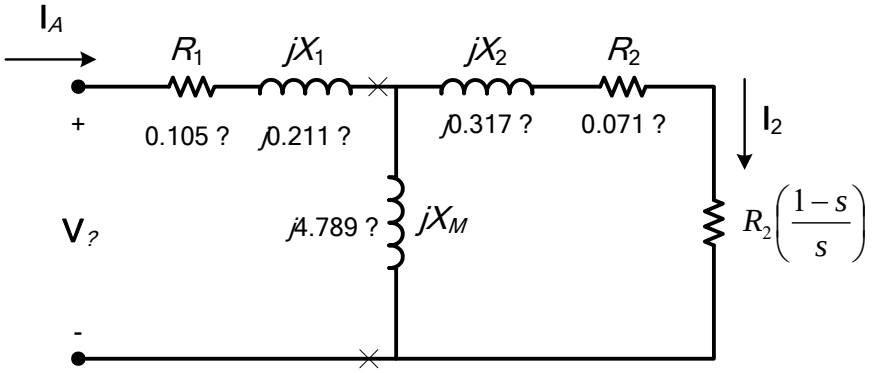
At a frequency of 60 Hz,

$$X_{\text{LR}} = \left(\frac{60\text{ Hz}}{15\text{ Hz}}\right) X'_{\text{LR}} = 0.528\ \Omega$$

For a Design Class B motor, the split is $X_1 = 0.211\ \Omega$ and $X_2 = 0.317\ \Omega$. Therefore,

$$X_M = 5.000\ \Omega - 0.211\ \Omega\ =\ 4.789\ \Omega$$

The resulting equivalent circuit is shown below:



6-21. A 460-V, 10 hp, four-pole, Y-connected, Insulation class F, Service Factor 1.15 induction motor has the following parameters

$R_1 = 0.54\ \Omega$ $R_2 = 0.488\ \Omega$ $X_M = 51.12\ \Omega$

$X_1 = 2.093\ \Omega$ $X_2 = 3.209\ \Omega$

$P_{F\&W} = 150$ W $P_{misc} = 50$ W $P_{core} = 150$ kW

For a slip of 0.02, find

(a) The line current I_L

(b) The stator power factor

(c) The rotor power factor

(d) The rotor frequency

(e) The stator copper losses P_{SCL}

(f) The air-gap power P_{AG}

(g) The power converted from electrical to mechanical form P_{conv}

(h) The induced torque τ_{ind}

(i) The load torque τ_{load}

(j) The overall machine efficiency η

(k) The motor speed in revolutions per minute and radians per second

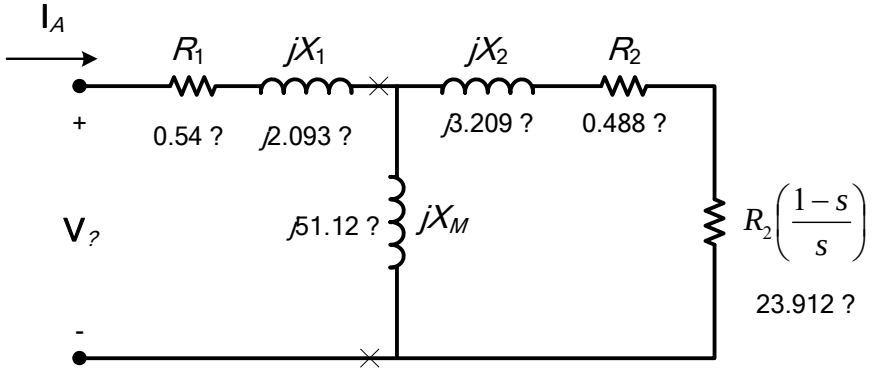
(l) Sketch the power flow diagram for this motor.

(m) What is the starting code letter for this motor?

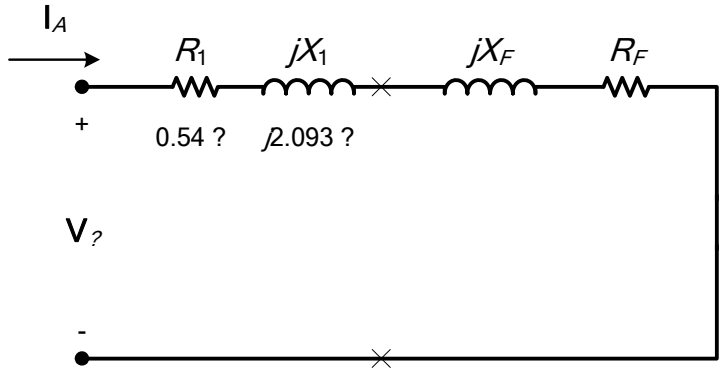
(n) What is the maximum acceptable temperature rise in this motor, given its insulation class?

(o) What does the service factor of this motor mean?

SOLUTION The equivalent circuit of this induction motor is shown below:



(a) The easiest way to find the line current (or armature current) is to get the equivalent impedance Z_F of the rotor circuit in parallel with jX_M, and then calculate the current as the phase voltage divided by the sum of the series impedances, as shown below.



The equivalent impedance of the rotor circuit in parallel with jX_M is:

$$Z_F = \frac{1}{\dfrac{1}{jX_M} + \dfrac{1}{Z_2}} = \frac{1}{\dfrac{1}{j51.12\ \Omega} + \dfrac{1}{24.41 + j3.209}} = 17.98 + j11.10 = 21.13\angle 31.7°\ \Omega$$

The phase voltage is $460/\sqrt{3} = 266$ V, so line current I_L is

$$I_L = I_A = \frac{V_\phi}{R_1 + jX_1 + R_F + jX_F} = \frac{266\angle 0°\text{ V}}{0.54\ \Omega + j2.093\ \Omega + 17.98\ \Omega + j11.10\ \Omega}$$

$$I_L = I_A = 11.70\angle -35.5°\text{ A}$$

(b) The stator power factor is

$$\text{PF} = \cos(35.5°) = 0.814 \text{ lagging}$$

(c) To find the rotor power factor, we must find the impedance angle of the rotor

$$\theta_R = \tan^{-1}\frac{X_2}{R_2/s} = \tan^{-1}\frac{3.209}{24.4} = 7.50°$$

(d) The rotor frequency is

$$f_r = sf_s = (0.02)(60\text{ Hz}) = 1.2\text{ Hz}$$

Therefore the rotor power factor is

$$\text{PF}_R = \cos 7.50° = 0.991 \text{ lagging}$$

(e) The stator copper losses are

$$P_{\text{SCL}} = 3I_A{}^2 R_1 = 3(11.7\text{ A})^2 (0.54\ \Omega) = 222\text{ W}$$

(f) The air gap power is $P_{\text{AG}} = 3I_2{}^2 \dfrac{R_2}{s} = 3I_A{}^2 R_F$

(Note that $3I_A{}^2 R_F$ is equal to $3I_2{}^2 \dfrac{R_2}{s}$, since the only resistance in the original rotor circuit was R_2/s, and the resistance in the Thevenin equivalent circuit is R_F. The power consumed by the Thevenin equivalent circuit must be the same as the power consumed by the original circuit.)

$$P_{\text{AG}} = 3I_2{}^2 \frac{R_2}{s} = 3I_A{}^2 R_F = 3(11.7\text{ A})^2 (17.98\ \Omega) = 7.38\text{ kW}$$

(g) The power converted from electrical to mechanical form is

$$P_{\text{conv}} = (1-s)P_{\text{AG}} = (1-0.02)\ (7.38\text{ kW}) = 7.23\text{ kW}$$

(h) The synchronous speed of this motor is

$$n_{\text{sync}} = \frac{120 f_{se}}{P} = \frac{120(60\text{ Hz})}{4} = 1800\text{ r/min}$$

$$\omega_{\text{sync}} = (1800\text{ r/min})\left(\frac{2\pi\text{ rad}}{1\text{ r}}\right)\left(\frac{1\text{ min}}{60\text{ s}}\right) = 188.5\text{ rad/s}$$

Therefore the induced torque in the motor is

$$\tau_{\text{ind}} = \frac{P_{\text{AG}}}{\omega_{\text{sync}}} = \frac{7.38\text{ kW}}{(1800\text{ r/min})\left(\frac{2\pi\text{ rad}}{1\text{ r}}\right)\left(\frac{1\text{ min}}{60\text{ s}}\right)} = 39.2\text{ N}\cdot\text{m}$$

(i) The output power of this motor is

$$P_{\text{OUT}} = P_{\text{conv}} - P_{\text{mech}} - P_{\text{core}} - P_{\text{misc}} = 7.23\text{ kW} - 150\text{ W} - 150\text{ W} - 50\text{ W} = 6.88\text{ kW}$$

The output speed is

$$n_m = (1-s)\ n_{\text{sync}} = (1-0.02)\ (1800\text{ r/min}) = 1764\text{ r/min}$$

Therefore the load torque is

$$\tau_{\text{load}} = \frac{P_{\text{OUT}}}{\omega_m} = \frac{6.88\text{ kW}}{(1764\text{ r/min})\left(\frac{2\pi\text{ rad}}{1\text{ r}}\right)\left(\frac{1\text{ min}}{60\text{ s}}\right)} = 37.2\text{ N}\cdot\text{m}$$

(j) The overall efficiency is

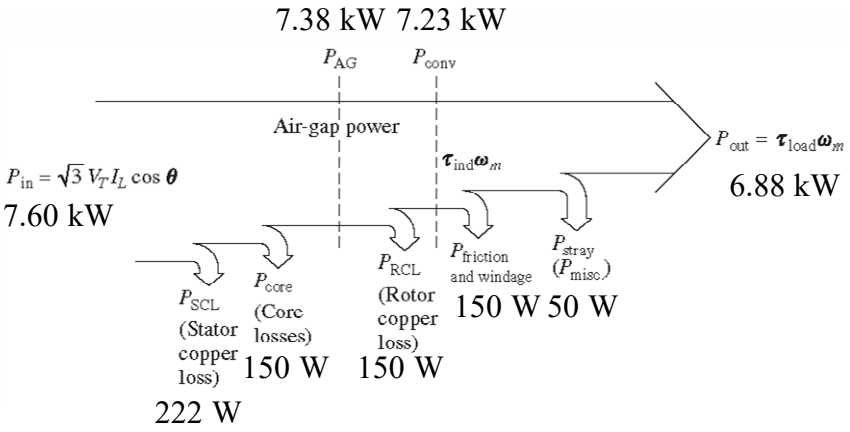
$$\eta = \frac{P_{\text{OUT}}}{P_{\text{IN}}} \times 100\% = \frac{P_{\text{OUT}}}{3V_\phi I_A \cos\theta} \times 100\%$$

$$\eta = \frac{6.88\text{ kW}}{3(266\text{ V})(11.7\text{ A})\cos(35.5°)} \times 100\% = 90.5\%$$

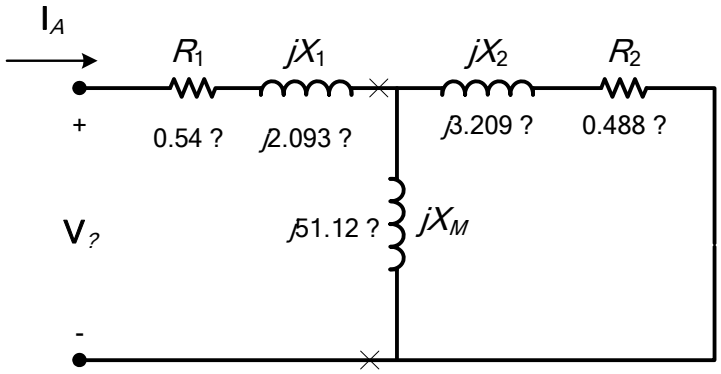
(k) The motor speed in revolutions per minute is 1764 r/min. The motor speed in radians per second is

$$\omega_m = (1764 \text{ r/min})\left(\frac{2\pi \text{ rad}}{1 \text{ r}}\right)\left(\frac{1 \text{ min}}{60 \text{ s}}\right) = 184.7 \text{ rad/s}$$

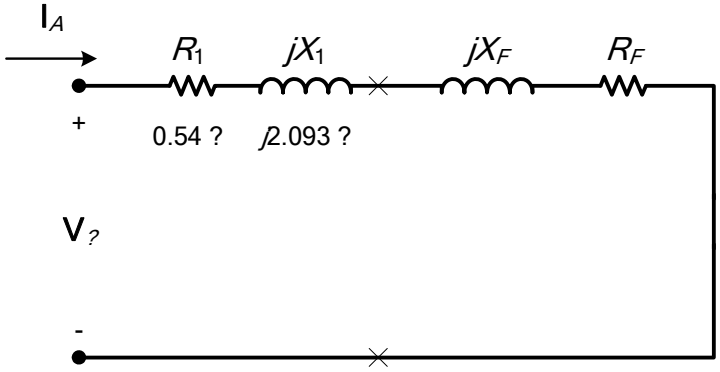
(l) The power flow diagram for this motor is



(m) The equivalent circuit of this induction motor *at starting conditions* is shown below:



The easiest way to find the line current (or armature current) is to get the equivalent impedance Z_F of the rotor circuit in parallel with jX_M, and then calculate the current as the phase voltage divided by the sum of the series impedances, as shown below.



The equivalent impedance of the rotor circuit in parallel with jX_M is:

$$Z_F = \frac{1}{\frac{1}{jX_M} + \frac{1}{Z_2}} = \frac{1}{\frac{1}{j51.12\ \Omega} + \frac{1}{0.488 + j3.209}} = 0.435 + j2.864 = 2.90\angle 81.3°\ \Omega$$

The phase voltage is $460/\sqrt{3} = 266$ V, so line current I_L is

$$I_L = I_A = \frac{V_\phi}{R_1 + jX_1 + R_F + jX_F} = \frac{266\angle 0° \text{ V}}{0.54\ \Omega + j2.093\ \Omega + 0.435\ \Omega + j2.90\ \Omega}$$

$$I_L = I_A = 52.3\angle -79° \text{ A}$$

The starting kVA of the motor is

$$S_{\text{start}} = 3V_\phi I_A = 3(266 \text{ V})(52.3 \text{ A}) = 41.7 \text{ kVA}$$

The locked rotor kVA/hp is

$$\text{kVA/hp} = \frac{41.7 \text{ kVA}}{10 \text{ hp}} = 4.17$$

Therefore this motor is Starting Code Letter D.

6-23. A 460-V four-pole 75-hp 60-Hz Y-connected three-phase induction motor develops its full-load induced torque at **3.5** percent slip when operating at 60 Hz and 460 V. The per-phase circuit model impedances of the motor are

$R_1 = 0.058\ \Omega$ $\quad$ $X_M = 18\ \Omega$

$X_1 = 0.32\ \Omega$ $\quad$ $X_2 = 0.386\ \Omega$

Mechanical, core, and stray losses may be neglected in this problem.

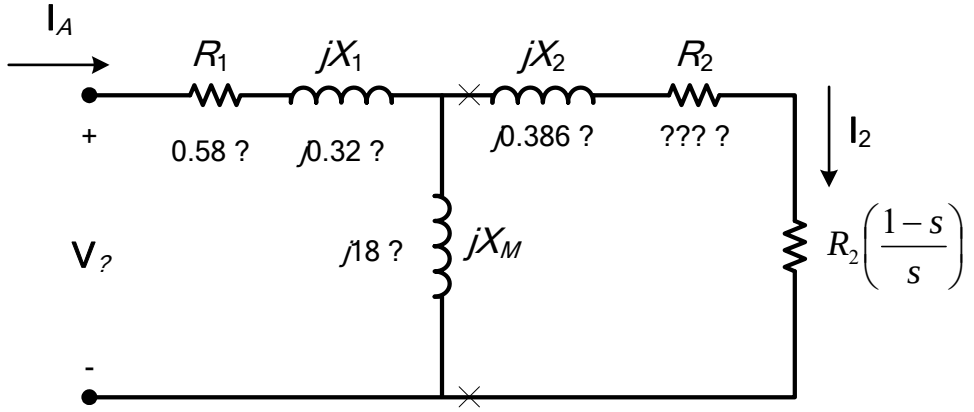
(a) Find the value of the rotor resistance R_2.

(b) Find $\tau_{\max}$, $s_{\max}$, and the rotor speed at maximum torque for this motor.

(c) Find the starting torque of this motor.

(d) What code letter factor should be assigned to this motor?

SOLUTION The equivalent circuit for this motor is



The Thevenin equivalent of the input circuit is:

$$Z_{\text{TH}} = \frac{jX_M\left(R_1 + jX_1\right)}{R_1 + j\left(X_1 + X_M\right)} = \frac{\left(j18\ \Omega\right)\left(0.58\ \Omega\ +\ j0.32\ \Omega\right)}{0.58\ \Omega\ +\ j\left(0.32\ \Omega + 18\ \Omega\right)} = 0.559 + j0.332\ \Omega = 0.651\angle 30.7°\ \Omega$$

$$\mathbf{V}_{\text{TH}} = \frac{jX_M}{R_1 + j\left(X_1 + X_M\right)}\mathbf{V}_\phi = \frac{\left(j18\ \Omega\right)}{0.58\ \Omega\ +\ j\left(0.32\ \Omega + 18\ \Omega\right)}\left(266\angle 0°\ \text{V}\right) = 261\angle 1.8°\ \text{V}$$

(a) If losses are neglected, the induced torque in a motor is equal to its load torque. At full load, the output power of this motor is 75 hp and its slip is 1.2%, so the induced torque is

$$n_m = \left(1 - 0.035\right)\left(1800\ \text{r/min}\right) = 1737\ \text{r/min}$$

$$\tau_{\text{ind}} = \tau_{\text{load}} = \frac{\left(75\ \text{hp}\right)\left(746\ \text{W/hp}\right)}{\left(1737\ \text{r/min}\right)\left(\frac{2\pi\ \text{rad}}{1\ \text{r}}\right)\left(\frac{1\,\text{min}}{60\ \text{s}}\right)} = 307.6\ \text{N}\cdot\text{m}$$

The induced torque is given by the equation

$$\tau_{\text{ind}} = \frac{3V_{\text{TH}}^2 R_2 / s}{\omega_{\text{sync}}\left[\left(R_{\text{TH}} + R_2/s\right)^2 + \left(X_{\text{TH}} + X_2\right)^2\right]}$$

Substituting known values and solving for R_2 / s yields

$$\tau_{\text{ind}} = \frac{3V_{\text{TH}}{}^2\left(R_2 / s\right)}{\omega_{\text{sync}}\left[\left(R_{\text{TH}} + R_2/s\right)^2 + \left(X_{\text{TH}} + X_2\right)^2\right]}$$

$$307.6\ \text{N}\cdot\text{m} = \frac{3\left(261\ \text{V}\right)^2 R_2 / s}{\left(188.5\ \text{rad/s}\right)\left[\left(0.559 + R_2/s\right)^2 + \left(0.332 + 0.386\right)^2\right]}$$

$$57{,}983 = \frac{205{,}932\ \ R_2 / s}{\left[\left(0.559 + R_2/s\right)^2 + 0.516\right]}$$

$$\left[\left(0.559 + R_2/s\right)^2 + 0.516\right] = 3.552\ \ R_2 / s$$

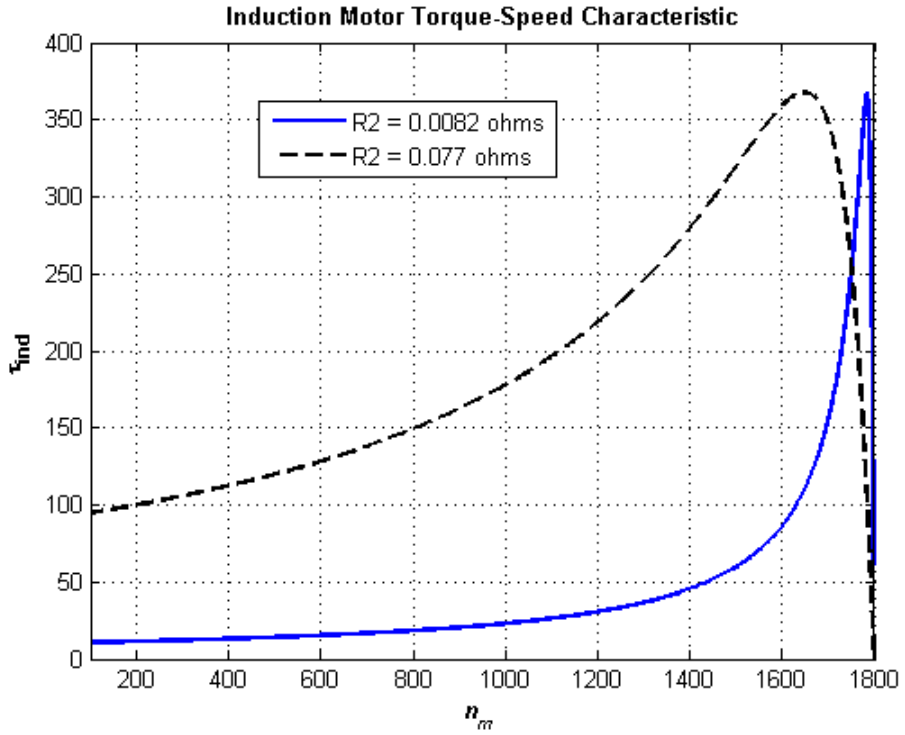
$$\left[0.3125 + 1.118R_2/s + \left(R_2/s\right)^2 + 0.516\right] = 3.552\ \ R_2 / s$$

$$\left(\frac{R_2}{s}\right)^2 - 2.434\left(\frac{R_2}{s}\right) + 0.516 = 0$$

$$\left(\frac{R_2}{s}\right) = 0.2346, \quad 2.199$$

$$R_2 = 0.0082\ \Omega,\ \ 0.077\ \Omega$$

These two solutions represent two situations in which the torque-speed curve would go through this specific torque-speed point. The two curves are plotted below. As you can see, only the 0.077 Ω solution is realistic, since the 0.0082 Ω solution passes through this torque-speed point at an unstable location on the back side of the torque-speed curve.



(b) The slip at pullout torque can be found by calculating the Thevenin equivalent of the input circuit from the rotor back to the power supply, and then using that with the rotor circuit model. The Thevenin equivalent of the input circuit was calculated in part *(a)*. The slip at pullout torque is

$$s_{\max} = \frac{R_2}{\sqrt{R_{\text{TH}}{}^2 + \left(X_{\text{TH}} + X_2\right)^2}}$$

$$s_{\max} = \frac{0.077\ \Omega}{\sqrt{\left(0.559\ \Omega\right)^2 + \left(0.332\ \Omega\ + 0.386\ \Omega\right)^2}} = 0.0846$$

The rotor speed a maximum torque is

$$n_{\text{pullout}} = (1 - s)\ n_{\text{sync}} = \left(1 - 0.0846\right)\left(1800\ \text{r/min}\right) = 1648\ \text{r/min}$$

and the pullout torque of the motor is

$$\tau_{\max} = \frac{3V_{\text{TH}}^2}{2\omega_{\text{sync}}\left[R_{\text{TH}} + \sqrt{R_{\text{TH}}^2 + \left(X_{\text{TH}} + X_2\right)^2}\right]}$$

$$\tau_{\max} = \frac{3(261\text{ V})^2}{2(188.5\text{ rad/s})\left[0.559\ \Omega + \sqrt{(0.559\ \Omega)^2 + (0.332\ \Omega\ + 0.368\ \Omega)^2}\right]}$$

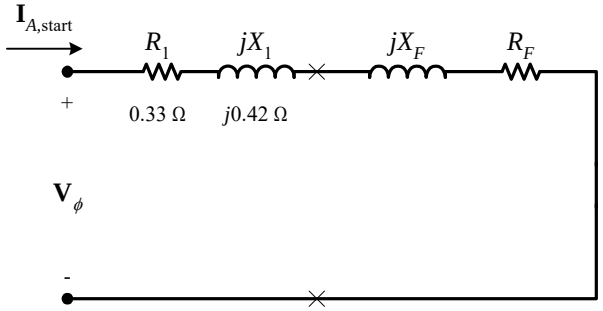
$$\tau_{\max} = 373\text{ N}\cdot\text{m}$$

(c) The starting torque of this motor is the torque at slip $s = 1$. It is

$$\tau_{\text{ind}} = \frac{3V_{\text{TH}}^2 R_2 / s}{\omega_{\text{sync}}\left[(R_{\text{TH}} + R_2/s)^2 + (X_{\text{TH}} + X_2)^2\right]}$$

$$\tau_{\text{ind}} = \frac{3(261\text{ V})^2(0.077\ \Omega)}{(188.5\text{ rad/s})\left[(0.559 + 0.077\ \Omega)^2 + (0.332 + 0.368)^2\right]} = 93.3\text{ N}\cdot\text{m}$$

(d) To determine the starting code letter, we must find the locked-rotor kVA per horsepower, which is equivalent to finding the starting kVA per horsepower. The easiest way to find the line current (or armature current) at starting is to get the equivalent impedance Z_F of the rotor circuit in parallel with jX_M at starting conditions, and then calculate the starting current as the phase voltage divided by the sum of the series impedances, as shown below.



The equivalent impedance of the rotor circuit in parallel with jX_M at starting conditions ($s = 1.0$) is:

$$Z_{F,\text{start}} = \frac{1}{\frac{1}{jX_M} + \frac{1}{Z_2}} = \frac{1}{\frac{1}{j18\ \Omega} + \frac{1}{0.077 + j0.386}} = 0.0838 + j0.3782 = 0.385\angle 79.0°\ \Omega$$

The phase voltage is $460/\sqrt{3}$ = 266 V, so line current $\mathbf{I}_{L,\text{start}}$ is

$$\mathbf{I}_{L,\text{start}} = \mathbf{I}_A = \frac{\mathbf{V}_\phi}{R_1 + jX_1 + R_F + jX_F} = \frac{266\angle 0°\text{ V}}{0.058\ \Omega + j0.32\ \Omega + 0.0838\ \Omega + j0.3782\ \Omega}$$

$$\mathbf{I}_{L,\text{start}} = \mathbf{I}_A = 373\angle -78.5°\text{ A}$$

Therefore, the locked-rotor kVA of this motor is

$$S = \sqrt{3}\ V_T\ I_{L,\text{rated}} = \sqrt{3}(460\text{ V})(373\text{ A}) = 297\text{ kVA}$$

and the kVA per horsepower is

$$\text{kVA/hp} = \frac{297\text{ kVA}}{75\text{ hp}} = 3.96\text{ kVA/hp}$$

This motor would have **starting code letter C**, since letter C covers the range 3.55 – 4.00.

6-24. Answer the following questions about the motor in Problem 6-21.

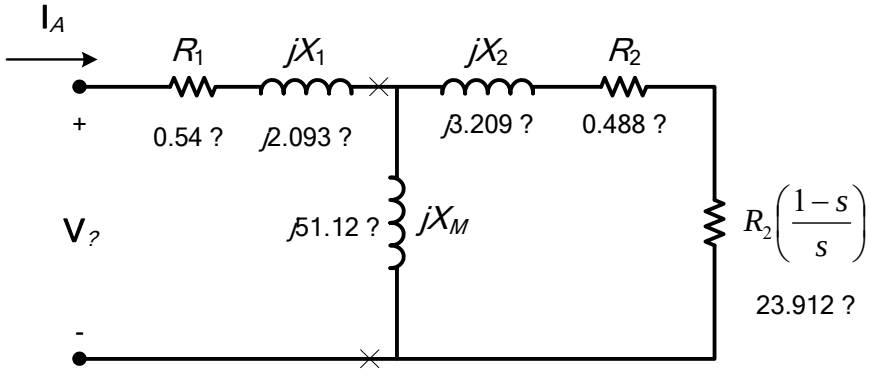
(a) If this motor is started from a 460-V infinite bus, how much current will flow in the motor at starting?

(b) If transmission line with an impedance of 0.50 + *j*0.35 Ω per phase is used to connect the induction motor to the infinite bus, what will the starting current of the motor be? What will the motor's terminal voltage be on starting?

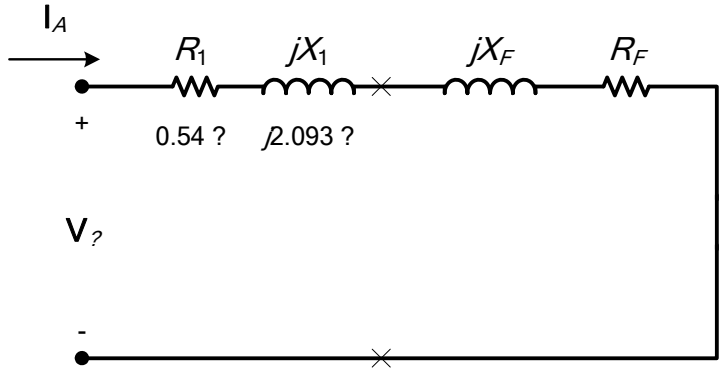
(c) If an ideal 1.4:1 step-down autotransformer is connected between the transmission line and the motor, what will the current be in the transmission line during starting? What will the voltage be at the motor end of the transmission line during starting?

SOLUTION

(a) The equivalent circuit of this induction motor is shown below:



The easiest way to find the line current (or armature current) is to get the equivalent impedance Z_F of the rotor circuit in parallel with jX_M, and then calculate the current as the phase voltage divided by the sum of the series impedances, as shown below.



The equivalent impedance of the rotor circuit in parallel with jX_M is:

$$Z_F = \frac{1}{\frac{1}{jX_M} + \frac{1}{Z_2}} = \frac{1}{\frac{1}{j51.12\ \Omega} + \frac{1}{0.488 + j3.209}} = 0.435 + j2.864 = 2.90\angle 81.3°\ \Omega$$

The phase voltage is $460/\sqrt{3} = 266$ V, so line current I_L is

$$\mathbf{I}_L = \mathbf{I}_A = \frac{\mathbf{V}_\phi}{R_1 + jX_1 + R_F + jX_F} = \frac{266\angle 0°\ \text{V}}{0.54\ \Omega + j2.093\ \Omega + 0.435\ \Omega + j2.90\ \Omega}$$

$$\mathbf{I}_L = \mathbf{I}_A = 52.3\angle -79°\ \text{A}$$

(b) If a transmission line with an impedance of 0.50 + *j*0.35 Ω per phase is used to connect the induction motor to the infinite bus, its impedance will be in series with the motor's impedances, and the starting current will be

$$\mathbf{I}_L = \mathbf{I}_A = \frac{\mathbf{V}_{\phi,\text{bus}}}{R_{\text{line}} + jX_{\text{line}} + R_1 + jX_1 + R_F + jX_F}$$

$$\mathbf{I}_L = \mathbf{I}_A = \frac{266\angle 0° \text{ V}}{0.50\ \Omega + j0.35\ \Omega + 0.54\ \Omega + j2.093\ \Omega + 0.435\ \Omega + j2.90\ \Omega}$$

$$\mathbf{I}_L = \mathbf{I}_A = 48.0\angle -74.6° \text{ A}$$

The voltage at the terminals of the motor will be

$$\mathbf{V}_\phi = \mathbf{I}_A\left(R_1 + jX_1 + R_F + jX_F\right)$$

$$\mathbf{V}_\phi = \left(48.0\angle -74.6° \text{ A}\right)\left(0.54\ \Omega + j2.093\ \Omega + 0.435\ \Omega + j2.90\ \Omega\right)$$

$$\mathbf{V}_\phi = 244.2\angle 4.3° \text{ V}$$

Therefore, the terminal voltage will be $\sqrt{3}\ \left(244.2 \text{ V}\right) = 423 \text{ V}$. Note that the terminal voltage sagged by about 9.2% during motor starting.

(c) If an ideal 1.4:1 step-down autotransformer is connected between the transmission line and the motor, the motor's impedances will be referred across the transformer by the square of the turns ratio *a* = 1.4. The referred impedances are

$$R_1' = a^2 R_1 = 1.96\ \left(0.058\ \Omega\right) = 0.1137\ \Omega$$

$$X_1' = a^2 X_1 = 1.96\ \left(0.32\ \Omega\right) = 0.627\ \Omega$$

$$R_F' = a^2 R_F = 1.96\ \left(0.435\ \Omega\right) = 0.853\ \Omega$$

$$X_F' = a^2 X_F = 1.96\ \left(2.90\ \Omega\right) = 5.684\ \Omega$$

Therefore, the starting current referred to the primary side of the transformer will be

$$\mathbf{I}_L' = \mathbf{I}_A' = \frac{\mathbf{V}_{\phi,\text{bus}}}{R_{\text{line}} + jX_{\text{line}} + R_1' + jX_1' + R_F' + jX_F'}$$

$$\mathbf{I}_L' = \mathbf{I}_A' = \frac{266\angle 0° \text{ V}}{0.50\ \Omega + j0.35\ \Omega + 0.1137\ \Omega + j0.627\ \Omega + 0.853\ \Omega + j5.684\ \Omega}$$

$$\mathbf{I}_L' = \mathbf{I}_A' = 39.0\angle -77.6° \text{ A}$$

The voltage at the motor end of the transmission line would be the same as the referred voltage at the terminals of the motor

$$\mathbf{V}_\phi' = \mathbf{I}_A'\left(R_1' + jX_1' + R_F' + jX_F'\right)$$

$$\mathbf{V}_\phi = \left(39.0\angle -77.6° \text{ A}\right)\left(0.1137\ \Omega + j0.627\ \Omega + 0.853\ \Omega + j5.684\ \Omega\right)$$

$$\mathbf{V}_\phi = 249\angle 3.7° \text{ V}$$

Therefore, the line voltage at the motor end of the transmission line will be $\sqrt{3}\ \left(249 \text{ V}\right) = 431 \text{ V}$. Note that this voltage sagged by 6.3% during motor starting, which is less than the 9.2% sag with case of across-the-line starting.

6-26. A 460-V 50-hp six-pole Δ-connected 60-Hz three-phase induction motor has a full-load slip of 4 percent, an efficiency of 91 percent, and a power factor of 0.87 lagging. At start-up, the motor develops 1.75 times the full-load torque but draws 7 times the rated current at the rated voltage. This motor is to be started with an autotransformer reduced voltage starter.

(a) What should the output voltage of the starter circuit be to reduce the starting torque until it equals the rated torque of the motor?

(b) What will the motor starting current and the current drawn from the supply be at this voltage?

SOLUTION

(a) The starting torque of an induction motor is proportional to the square of V_{TH},

$$\frac{\tau_{start2}}{\tau_{start1}} = \left(\frac{V_{TH2}}{V_{TH1}}\right)^2 = \left(\frac{V_{T2}}{V_{T1}}\right)^2$$

If a torque of 1.75 τ_{rated} is produced by a voltage of 460 V, then a torque of 1.00 τ_{rated} would be produced by a voltage of

$$\frac{1.00\ \tau_{rated}}{1.75\ \tau_{rated}} = \left(\frac{V_{T2}}{460\ \text{V}}\right)^2$$

$$V_{T2} = \sqrt{\frac{(460\ \text{V})^2}{1.75}} = 348\ \text{V}$$

(b) The motor starting current is directly proportional to the starting voltage, so

$$I_{L2} = \left(\frac{348\ \text{V}}{460\ \text{V}}\right) I_{L1} = (0.757) I_{L1} = (0.757)(7 I_{rated}) = 5.30\ I_{rated}$$

The input power to this motor is

$$P_{IN} = \frac{P_{OUT}}{\eta} = \frac{(50\ \text{hp})(746\ \text{W/hp})}{0.91} = 41.0\ \text{kW}$$

The rated current is equal to

$$I_{rated} = \frac{P_{IN}}{\sqrt{3}\ V_T\ \text{PF}} = \frac{(41.0\ \text{kW})}{\sqrt{3}(460\ \text{V})(0.87)} = 59.15\ \text{A}$$

Therefore, the motor starting current is

$$I_{L2} = 5.445\ I_{rated} = (5.30)(59.15\ \text{A}) = 313.5\ \text{A}$$

The turns ratio of the autotransformer that produces this starting voltage is

$$\frac{N_{SE} + N_C}{N_C} = \frac{460\ \text{V}}{348\ \text{V}} = 1.322$$

so the current drawn from the supply will be

$$I_{line} = \frac{I_{start}}{1.377} = \frac{313.5\ \text{A}}{1.322} = 237\ \text{A}$$

6-27. A wound-rotor induction motor is operating at rated voltage and frequency with its slip rings shorted and with a load of about 25 percent of the rated value for the machine. If the rotor resistance of this machine is doubled by inserting external resistors into the rotor circuit, explain what happens to the following:

(a) Slip s

(b) Motor speed n_m

(c) The induced voltage in the rotor

(d) The rotor current

(e) τ_{ind}

(f) P_{out}

(g) P_{RCL}

(h) Overall efficiency η

SOLUTION

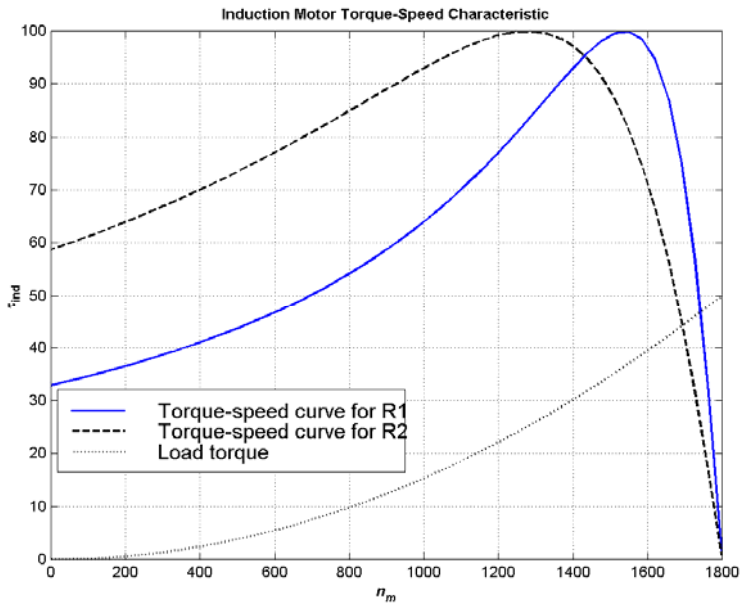
(a) The slip s will increase.

(b) The motor speed n_m will decrease.

(c) The induced voltage in the rotor will increase.

(d) The rotor current will increase.

(e) The induced torque will adjust to supply the load's torque requirements at the new speed. This will depend on the shape of the load's torque-speed characteristic. For most loads, the induced torque will decrease.



(f) The output power will generally decrease: $P_{\text{OUT}} = \tau_{\text{ind}} \downarrow \omega_m \downarrow$

(g) The rotor copper losses (including the external resistor) will increase.

(h) The overall efficiency η will decrease.

6-28. A 460-V, 75 hp, four-pole, Y-connected induction motor has the following parameters

$R_1 = 0.058\ \Omega$ $R_2 = 0.037\ \Omega$ $X_M = 9.24\ \Omega$

$X_1 = 0.320\ \Omega$ $X_2 = 0.386\ \Omega$

$P_{F\&W} = 650$ W $P_{misc} = 150$ W $P_{core} = 600$ kW

For a slip of 0.01, find

(a) The line current I_L

(b) The stator power factor

(c) The rotor power factor

(d) The rotor frequency

(e) The stator copper losses P_{SCL}

(f) The air-gap power P_{AG}

(g) The power converted from electrical to mechanical form P_{conv}

(h) The induced torque τ_{ind}

(i) The load torque τ_{load}

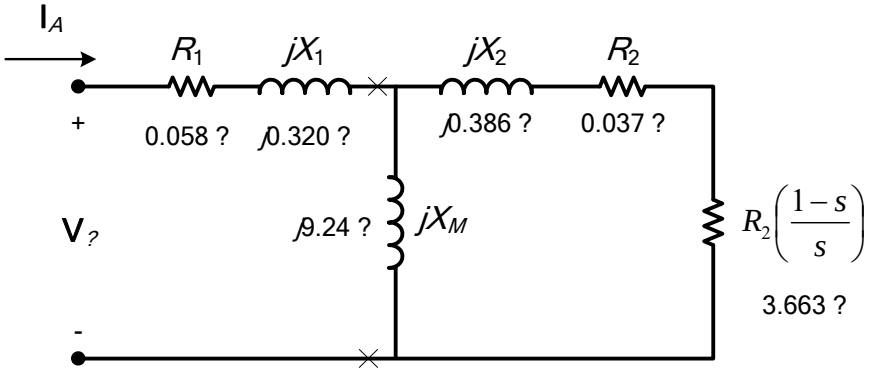
(j) The overall machine efficiency η

(k) The motor speed in revolutions per minute and radians per second

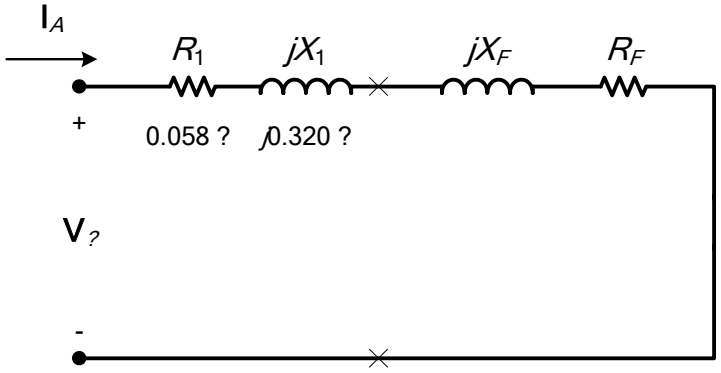
(l) Sketch the power flow diagram for this motor.

(m) What is the starting code letter for this motor?

SOLUTION The equivalent circuit of this induction motor is shown below:



(a) The easiest way to find the line current (or armature current) is to get the equivalent impedance Z_F of the rotor circuit in parallel with jX_M, and then calculate the current as the phase voltage divided by the sum of the series impedances, as shown below.



The equivalent impedance of the rotor circuit in parallel with jX_M is:

$$Z_F = \frac{1}{\frac{1}{jX_M} + \frac{1}{Z_2}} = \frac{1}{\frac{1}{j9.24\ \Omega} + \frac{1}{3.70 + j0.386}} = 2.970 + j1.512 = 3.333\angle 27.0°\ \Omega$$

The phase voltage is $460/\sqrt{3} = 266$ V, so line current I_L is

$$\mathbf{I}_L = \mathbf{I}_A = \frac{\mathbf{V}_\phi}{R_1 + jX_1 + R_F + jX_F} = \frac{266\angle 0°\ \text{V}}{0.058\ \Omega + j0.32\ \Omega + 2.97\ \Omega + j1.512\ \Omega}$$

$$\mathbf{I}_L = \mathbf{I}_A = 75.2\angle -31.2°\ \text{A}$$

(b) The stator power factor is

$$\text{PF} = \cos(31.2°) = 0.855 \text{ lagging}$$

(c) To find the rotor power factor, we must find the impedance angle of the rotor

$$\theta_R = \tan^{-1}\frac{X_2}{R_2/s} = \tan^{-1}\frac{0.386}{0.037/0.01} = 5.96°$$

(d) The rotor frequency is

$$f_r = sf_s = (0.01)(60\ \text{Hz}) = 0.6\ \text{Hz}$$

Therefore the rotor power factor is

$$\text{PF}_R = \cos 5.96° = 0.995 \text{ lagging}$$

(e) The stator copper losses are

$$P_{\text{SCL}} = 3I_A^2 R_1 = 3(75.2\ \text{A})^2(0.058\ \Omega) = 984\ \text{W}$$

(f) The air gap power is $P_{\text{AG}} = 3I_2^2\frac{R_2}{s} = 3I_A^2 R_F$

(Note that $3I_A^2 R_F$ is equal to $3I_2^2\frac{R_2}{s}$, since the only resistance in the original rotor circuit was R_2/s, and the resistance in the Thevenin equivalent circuit is R_F. The power consumed by the Thevenin equivalent circuit must be the same as the power consumed by the original circuit.)

$$P_{\text{AG}} = 3I_2^2\frac{R_2}{s} = 3I_A^2 R_F = 3(75.2\ \text{A})^2(2.97\ \Omega) = 50.4\ \text{kW}$$

(g) The power converted from electrical to mechanical form is

$$P_{\text{conv}} = (1-s)P_{\text{AG}} = (1-0.01)\ (50.4\ \text{kW}) = 49.9\ \text{kW}$$

(h) The synchronous speed of this motor is

$$n_{\text{sync}} = \frac{120\, f_{se}}{P} = \frac{120(60\text{ Hz})}{4} = 1800\text{ r/min}$$

$$\omega_{\text{sync}} = (1800\text{ r/min})\left(\frac{2\pi\text{ rad}}{1\text{ r}}\right)\left(\frac{1\text{ min}}{60\text{ s}}\right) = 188.5\text{ rad/s}$$

Therefore the induced torque in the motor is

$$\tau_{\text{ind}} = \frac{P_{\text{AG}}}{\omega_{\text{sync}}} = \frac{50.4\text{ kW}}{(1800\text{ r/min})\left(\frac{2\pi\text{ rad}}{1\text{ r}}\right)\left(\frac{1\text{ min}}{60\text{ s}}\right)} = 267.4\text{ N}\cdot\text{m}$$

(i) The output power of this motor is

$$P_{\text{OUT}} = P_{\text{conv}} - P_{\text{mech}} - P_{\text{core}} - P_{\text{misc}} = 49.9\text{ kW} - 650\text{ W} - 600\text{ W} - 150\text{ W} = 48.5\text{ kW}$$

The output speed is

$$n_m = (1-s)\ n_{\text{sync}} = (1-0.01)\ (1800\text{ r/min}) = 1782\text{ r/min}$$

Therefore the load torque is

$$\tau_{\text{load}} = \frac{P_{\text{OUT}}}{\omega_m} = \frac{48.5\text{ kW}}{(1782\text{ r/min})\left(\frac{2\pi\text{ rad}}{1\text{ r}}\right)\left(\frac{1\text{ min}}{60\text{ s}}\right)} = 260\text{ N}\cdot\text{m}$$

(j) The overall efficiency is

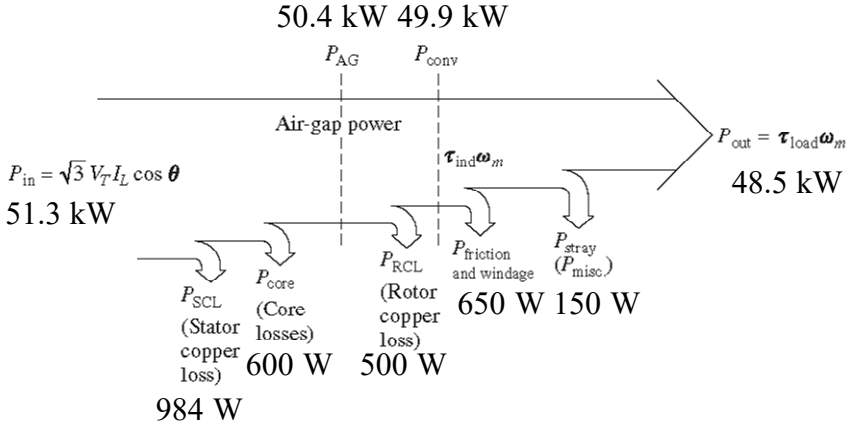
$$\eta = \frac{P_{\text{OUT}}}{P_{\text{IN}}} \times 100\% = \frac{P_{\text{OUT}}}{3V_\phi I_A \cos\theta} \times 100\%$$

$$\eta = \frac{48.5\text{ kW}}{3(266\text{ V})(75.2\text{ A})\cos(31.2°)} \times 100\% = 94.5\%$$

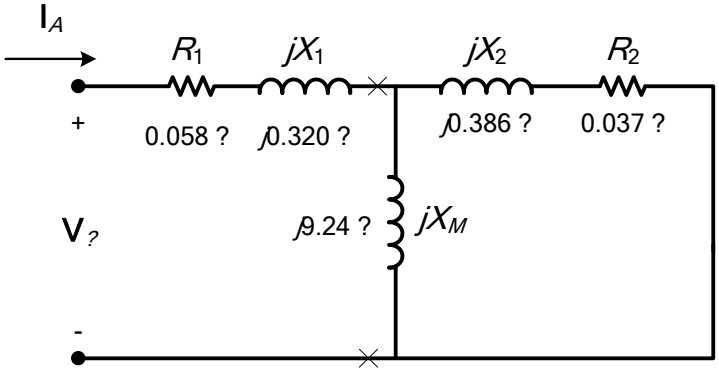
(k) The motor speed in revolutions per minute is 1782 r/min. The motor speed in radians per second is

$$\omega_m = (1782\text{ r/min})\left(\frac{2\pi\text{ rad}}{1\text{ r}}\right)\left(\frac{1\text{ min}}{60\text{ s}}\right) = 186.6\text{ rad/s}$$

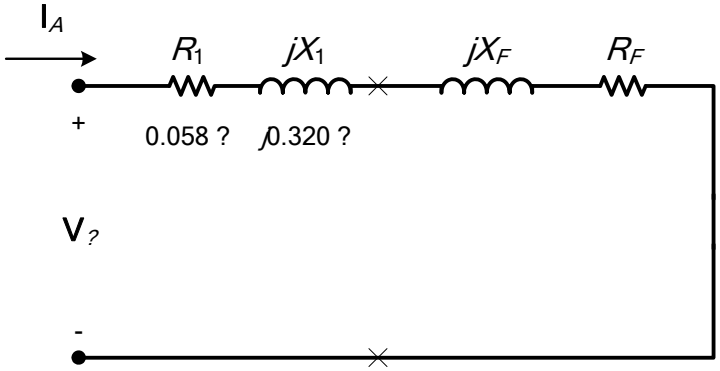
(l) The power flow diagram for this motor is



(m) The equivalent circuit of this induction motor *at starting conditions* is shown below:



The easiest way to find the line current (or armature current) is to get the equivalent impedance Z_F of the rotor circuit in parallel with jX_M, and then calculate the current as the phase voltage divided by the sum of the series impedances, as shown below.



The equivalent impedance of the rotor circuit in parallel with jX_M is:

$$Z_F = \frac{1}{\frac{1}{jX_M}+\frac{1}{Z_2}} = \frac{1}{\frac{1}{j9.24\ \Omega}+\frac{1}{0.037+j0.386}} = 0.0341+j0.3707 = 0.372\angle 84.7^\circ\ \Omega$$

The phase voltage is $460/\sqrt{3} = 266$ V, so line current I_L is

$$\mathbf{I}_L = \mathbf{I}_A = \frac{\mathbf{V}_\phi}{R_1 + jX_1 + R_F + jX_F} = \frac{266\angle 0^\circ\ \text{V}}{0.058\ \Omega\ + j0.32\ \Omega + 0.0341\ \Omega + j0.3707\ \Omega}$$

$$\mathbf{I}_L = \mathbf{I}_A = 381.7\angle -82.4^\circ\ \text{A}$$

The starting kVA of the motor is

$$S_{\text{start}} = 3V_\phi I_A = 3\left(266\ \text{V}\right)\left(381.7\ \text{A}\right) = 304.6\ \text{kVA}$$

The locked rotor kVA/hp is

$$\text{kVA/hp} = \frac{304.6\ \text{kVA}}{75\ \text{hp}} = 4.06$$

Therefore this motor is Starting Code Letter D.

6-31. When it is necessary to stop an induction motor very rapidly, many induction motor controllers reverse the direction of rotation of the magnetic fields by switching any two stator leads. When the direction of rotation of the magnetic fields is reversed, the motor develops an induced torque opposite to the current direction of rotation, so it quickly stops and tries to start turning in the opposite direction. If power is removed from the stator circuit at the moment when the rotor speed goes through zero, then the motor has been stopped very rapidly. This technique for rapidly stopping an induction motor is called *plugging*. The motor of Problem **6-21** is running at rated conditions and is to be stopped by plugging.

(a) What is the slip *s* before plugging?

(b) What is the frequency of the rotor before plugging?

(c) What is the induced torque τ_{ind} before plugging?

(d) What is the slip *s* immediately after switching the stator leads?

(e) What is the frequency of the rotor immediately after switching the stator leads?

(f) What is the induced torque τ_{ind} immediately after switching the stator leads?

SOLUTION

(a) The slip before plugging is 0.02 (see Problem 6-21).

(b) The frequency of the rotor before plugging is $f_r = sf_e = (0.02)(60\text{ Hz}) = 1.2\text{ Hz}$

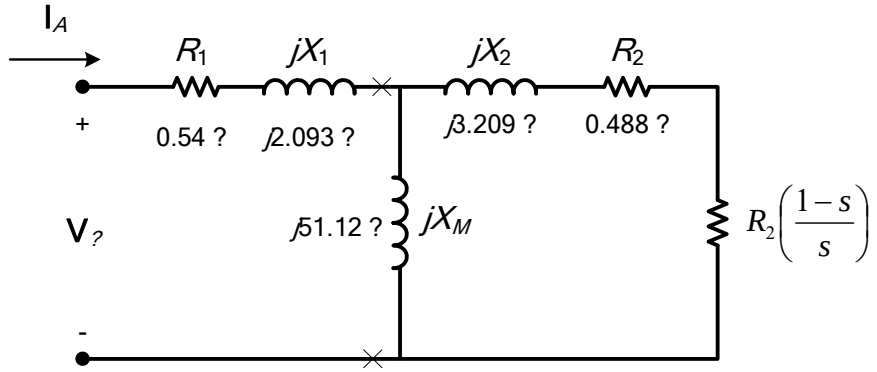
(c) The induced torque before plugging is 39.2 N·m in the direction of motion (see Problem 6-21).

(d) After switching stator leads, the synchronous speed becomes –1800 r/min, while the mechanical speed initially remains 1764 r/min. Therefore, the slip becomes

$$s = \frac{n_{\text{sync}} - n_m}{n_{\text{sync}}} = \frac{-1800 - 1764}{-1800} = 1.98$$

(e) The frequency of the rotor after plugging is $f_r = sf_e = (1.98)(60\text{ Hz}) = 118.8\text{ Hz}$

(f) The equivalent circuit for this motor is



The Thevenin equivalent of the input circuit is:

$$Z_{\text{TH}} = \frac{jX_M(R_1 + jX_1)}{R_1 + j(X_1 + X_M)} = \frac{(j51.12\ \Omega)(0.54\ \Omega + j2.093\ \Omega)}{0.54\ \Omega + j(2.093\ \Omega + 51.12\ \Omega)} = 0.4983 + j2.016\ \Omega = 2.076\angle 76.1°\ \Omega$$

$$\mathbf{V}_{\text{TH}} = \frac{jX_M}{R_1 + j(X_1 + X_M)}\mathbf{V}_\phi = \frac{(j51.2\ \Omega)}{0.54\ \Omega + j(2.093\ \Omega + 51.2\ \Omega)}(266\angle 0°\text{ V}) = 256\angle 0.58°\text{ V}$$

The induced torque immediately after switching the stator leads is

$$\tau_{\text{ind}} = \frac{3V_{\text{TH}}^2 R_2 / s}{\omega_{\text{sync}}\left[(R_{\text{TH}} + R_2/s)^2 + (X_{\text{TH}} + X_2)^2\right]}$$

$$\tau_{\text{ind}} = \frac{3(256\text{ V})^2(0.488\ \Omega / 1.962)}{(188.5\text{ rad/s})\left[(0.4983 + 0.488\ \Omega/1.962)^2 + (2.016 + 3.209)^2\right]}$$

$$\tau_{\text{ind}} = \frac{3(256\text{ V})^2(0.2487\ \Omega)}{(188.5\text{ rad/s})\left[(0.4983 + 0.2487\ \Omega)^2 + (2.016 + 3.209)^2\right]}$$

$\tau_{\text{ind}} = 9.3\text{ N}\cdot\text{m}$, opposite the direction of motion

6-32. A 460-V, 10 hp, two-pole, Y-connected induction motor has the following parameters

$R_1 = 0.54\ \Omega$ $X_1 = 2.093\ \Omega$ $X_M = 51.12\ \Omega$

$P_{F\&W} = 150$ W $P_{misc} = 50$ W $P_{core} = 150$ kW

The rotor is a dual-cage design, with a tightly-coupled, high resistance outer bar and a loosely-coupled, low resistance inner bar (see Figure 6-25*c*). The parameters of the outer bar are

$R_{2o} = 4.80\ \Omega$ $X_{2o} = 3.75\ \Omega$

The resistance is high due to the lower cross sectional area, and the reactance is relatively low due to the tight coupling between the rotor and stator. The parameters of the inner bar are

$R_{2i} = 0.573\ \Omega$ $X_{2i} = 4.65\ \Omega$

The resistance is low due to the high cross sectional area, but the reactance is relatively high due to the quite loose coupling between the rotor and stator.

Calculate the torque-speed characteristic for this induction motor, and compare it to the torque-speed characteristic for the single-cage design in Problem 6-21. How do the curves differ? Explain the differences.

SOLUTION The dual-cage rotor has two current paths in parallel, the inner cage and the outer cage. As a result, the impedance of the rotor is calculated as the parallel combination of these two current paths. For any given slip, the impedance of the rotor can be calculated as

$$Z_R = \frac{1}{\frac{1}{R_i + jX_i} + \frac{1}{R_o + jX_o}}$$

where R_i is the resistance of the inner rotor cage, X_i is the reactance of the inner rotor cage, and so forth. Also, recall that rotor reactance varies with rotor frequency. The rotor reactance is given by the equation

$$X = sX_o$$

where *s* is the slip and X_o is the rotor reactance at locked-rotor conditions. The rotor impedance and any slip can thus be calculated as

$$Z_R = \frac{1}{\frac{1}{R_i + jsX_{oi}} + \frac{1}{R_o + jsX_{oo}}}$$

where X_{oi} is the reactance of the inner rotor cage at locked-rotor conditions, and X_{oo} is the reactance of the outer rotor cage at locked-rotor conditions. We must apply this equation to calculate the rotor impedance at any slip, and then divide the resulting reactance by the slip to get to the equivalent impedance at locked-rotor conditions (the reactance at locked-rotor conditions is the term that goes into the torque equation).